

QUESTION OF THE WEEK 5

GRADE XI CHEMISTRY

Assignment Submission Date: Bhadra 1

1. Differentiate between electron affinity and electronegativity.
2. Compare the size of S^{--} , Cl^- , Ar , K^+ and Ca^{++} with reason.
3. Define electron affinity. What are the various factors that affect electron affinity?
4. State and explain Graham's Law of Diffusion. If the time of equal effusion of pure oxygen and a mixture of oxygen and ozone are 224 sec and 243 sec respectively, find the percentage of ozone in the mixture.
5. State and explain Dalton's Law of partial pressure. 500 cc of oxygen under 680 mm pressure and 250 cc of nitrogen at 720 mm pressure were put together in a litre flask keeping the temperature constant. Find the final pressure of the mixture.
6. Balance the following redox reaction by ion-electron method.
 - a) $Cl_2 + OH^- \longrightarrow Cl^- + ClO_3^- + H_2O$
 - b) $MnO_4^- + H_2O_2 + H^+ \longrightarrow MnO_2 + O_2 + H_2O$

1. Find from definition the derivatives of the following functions:

a) $\sin(3 - x)$

b) $\sqrt{\tan x}$

c) $\sqrt{\cot x}$

d) $\sec(ax + b)$

2. Find $\frac{dy}{dx}$ when:

a) $(x + y)^5 = x^3 y^2$

b) $\sqrt{x} + \sqrt{y} = \sqrt{xy}$

c) $x^2 + y^2 = (x + y)^2$

d) $\sin(x + y) + \sin(x - y) = 1$

e) $\cos(x + y) = \tan xy$

3. Find $\frac{dy}{dx}$ when:

a) $y = ax^2 + bt + c$ and $t = \sqrt{x^2 + 1}$

b) $y = 3\theta^2 + 4\theta + 5$ and $\theta = 3x^2 + 4x + 5$

c) $y = at^3$ and $x = bt^2$

d) $y = \frac{a}{t}$ and $x = \frac{b}{t^3}$

Weekly question week 5

Grade XI Physics

Assignment Submission Date: Bhadra 1

- a. If $\vec{A}$ and $\vec{B}$ are non-zero vectors, is it possible $\vec{A} \times \vec{B}$ and $\vec{A} \cdot \vec{B}$ both to be zero at the same time? Explain.
- b. What do you mean by resolution of a vector?
- c. Discuss in detail, the resolution of a vector in two dimensions.
- d. Explain the geometrical interpretation of scalar product of two vectors.
- e. Vectors; $\vec{A} = 3\hat{i} + 5\hat{j} - 2\hat{k}$ and $\vec{B} = -3\hat{j} + 6\hat{k}$. Find the vector $\vec{C}$, such that $2\vec{A} + 7\vec{B} + 4\vec{C} = 0$.
- f. Find the scalar product of the two vectors: $\vec{A} = 4.00\hat{i} + 3.00\hat{j}$ and $\vec{B} = 5.00\hat{i} + 2.00\hat{j}$. Also, find the angle between them.
- g. a. Two spheres of same radius are made up of same material but one is hollow and the other is solid. Will their overall expansion be same or different if they are heated from 20°C to 30°C ? Explain?
- h. A Steel rod and a brass rod when measured at 0°C are 50.1 cm and 50cm respectively. At what temperature will there length become equal and what will be their common length? Given, $\alpha_{\text{steel}} = 12 \times 10^{-6}/^\circ\text{C}$ and $\alpha_{\text{brass}} = 18 \times 10^{-6}/^\circ\text{C}$.
- i. Does the density of a substance depend on temperature? Explain.
- j. Derive an expression showing the variation of density of a substance with the change in temperature.
- k. The coefficient of real expansion of a substance is $1.6 \times 10^{-4}/^\circ\text{C}$. If the density of the substance is 10 g/cc at 0°C , find its density at 673K .
- l. On heating a glass block of 10,000 cm^3 , from 25°C to 40°C , its volume increases by 4cm^3 . Calculate coefficient of linear expansion of the glass.
- m. The difference between the lengths of a certain brass rod and that of a steel rod is claimed to be constant at all temperatures. Is this possible? Explain.
- n. Describe the experiment method for the determination of linear expansivity of a material in the form of a rod.
- o. A glass rod rubbed with silk is found to have acquired a positive charge of $6.4 \times 10^{-7}\text{C}$. Does the glass rod lose any mass in this process? If so, estimate the loss of mass.
- p. State and explain Coulomb's law in electrostatics.
- q. Point Charges having values $+5\mu\text{C}$, $+10\mu\text{C}$, and $-10\mu\text{C}$ are placed in corners A, B and C of an equilateral triangle ABC having each side equal to 5 cm. Find the resultant force on the charge at A.
- r. State and explain the principle of superposition of Coulomb's force.
- s. Why coulomb's law is applicable to stationary charges only?
- t. Two point charges of $10\mu\text{C}$ and $-90\mu\text{C}$ are separated by a distance of 18 cm. Find the point along the line joining them at which electric field intensity is zero.

QUESTION OF THE WEEK (WEEK 4)

CLASS XI (sub: PHYSICS)

- 1) a) What are the limitations of dimensional analysis?
 b) Check the correctness of the physical relation $\rho = \frac{3g}{4\pi r G}$ where symbols carry their usual meanings.
 c) Find the value of 20 J on the system based on 20 cm, 1kg and $\frac{1}{2}$ minute as fundamental units.
- 2) a) State and use parallelogram law of vector addition to find the magnitude and direction of the resultant of two vectors inclined at an angle θ .
 b) Two forces F_1 and F_2 act at a point. F_1 has magnitude 8N and acts at an angle of 30° with the positive x-axis. F_2 has a magnitude of 6N and acts at an angle of 150° with positive x-axis. Find the magnitude and direction of resultant force.
- 3) a) Define the coefficient linear, superficial and cubical expansions. Derive relations between them.
 b) Why a thick glass cracks when boiling water is poured in it?
- 4) a) Pendulum clocks generally go fast in winter and slow in summer. Why?
 b) A pendulum clock loses 12 Sec. a day if the temperature is 40°C and gains 4 Sec. a day if the temperature is 20°C . Find the temperature at which the clock will show correct time and coefficient of linear expansion of the metal of pendulum shaft.
- 5) a) State and explain the principle of superposition of Coulomb's force.
 b) Two-point charges of $10\ \mu\text{C}$ and $-40\ \mu\text{C}$ are separated by distance of 13 cm. Find the point along the line joining them at which net electrostatic force on any charge is zero.
- 6) a) Define dielectric constant. Why it's value is always greater than one?
 b) Four equal charges q are placed at the four corners of a square of length a . Find the magnitude of the net force acting on one of them due to other charges and draw diagram to show the direction of force on that charge.
- 7) a) Explain why vector addition is commutative, providing a real-life example to support your reasoning.
 b) What is meant by resolution of vectors? How will you resolve a vector in a plane in two mutually perpendicular directions?
- 8) a) Why does a solid expand on heating?
 b) Explain an experiment to determine the coefficient of linear expansion of a solid.
 c) A steel cylinder of a car engine has an aluminium piston. The internal diameter of the cylinder at 15°C is 8.0 cm and there is an all-round clearance of 0.05 mm between the piston and the cylinder of the wall. Find the temperature at which the two will exactly fit in. ($\alpha_{\text{steel}} = 1.2 \times 10^{-5}\ ^\circ\text{C}^{-1}$ and $\alpha_{\text{aluminium}} = 1.6 \times 10^{-5}\ ^\circ\text{C}^{-1}$)
- 9) a) State and explain Coulomb's law in electrostatics.
 b) Why coulomb's law is applicable to stationary charges only?
 c) Define 1 coulomb charge on the basis of Coulomb's law.
 d) Two charges 10^{-6}C and 10^{-7}C attract each other with a force of 400N. Calculate the distance between them if they are in i) air and ii) water (relative permittivity of water is 80)

QUESTION OF THE WEEK 5

GRADE XI CHEMISTRY

Assignment Submission Date: Bhadra 1

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Weekly question week 5

Grade XI Physics

Assignment Submission Date: Bhadra 1

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- f. Find the scalar product of the two vectors: $\vec{A} = 4.00\hat{i} + 3.00\hat{j}$ and $\vec{B} = 5.00\hat{i} + 2.00\hat{j}$. Also, find the angle between them.
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FIRST TERM EXAMINATION 2082

Class: XI 'Science'

SET: 'B'

Time: 25 Min.

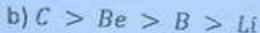
Subject: Chemistry (3011)

Write the best options among the four options on your answer sheet.

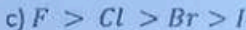
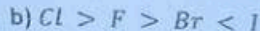
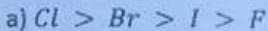
GROUP - 'A' (MCQ) [11×1 = 11]

1. 3.7 g of a gas at 25° C occupy the same volume as 0.184 g of H_2 at 17° C and at same pressure. What is the molecular mass of gas?
 a) 40.21 amu b) 45.35 amu c) 41.33 amu d) 46.32 amu
2. Diffusion of He is
 a) 4 times faster than CO_2 b) 4 times faster than SO_2
 c) 4 times faster than NO_2 d) 4 times faster than ClO_2
3. The temperature below which gas does not obey ideal gas law is
 a) Critical temperature b) Inversion temperature
 c) Boyle's temperature d) Reduced temperature
4. 300 ml of a gas at 27° C is cooled to - 3° C at constant pressure. The final volume of is
 a) 540 ml b) 270 ml c) 135 ml d) 350 ml
5. An element M has an atomic mass 19 and the atomic number 9. It's ion is represented by
 a) M^+ b) M^{2-} c) M^- d) M^{2+}
6. The oxidation number of sulphur in S_8 , S_2F_2 and H_2S are respectively
 a) 0, +1, -2 b) +2, +1, -2 c) 0, +1, +2 d) -2, +1, -2
7. Of the following reactions, which involves redox process?
 a) $HI + AgNO_3 \longrightarrow AgI + HNO_3$
 b) $HCl + NaBr \longrightarrow NaCl + HBr$
 c) $HCl + NaOH \longrightarrow NaCl + H_2O$
 d) $H_2 + Br_2 \longrightarrow 2 HBr$

8. The correct order of decreasing first ionization energy is



9. Which of the following represents the correct order of electron affinities?



10. Which is the best reducing agent?



11. Which one of the following ions has the smallest radius?



KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2082

Class: XI 'Science'

Time: 3hrs.

Subject: Chemistry [3011]

F.M.: 75

P.M.: 30

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

GROUP - 'B' [8x5=40]

SHORT ANSWER QUESTIONS.

1. a) Define electron affinity. [1]
b) Giving reasons, explain the variation of electron affinity in a period and group. [2]
c) Be and Mg have almost zero electron affinity, why? [2]
2. a) State modern periodic law. [1]
b) Discuss the various features of modern periodic table. [4]
3. a) Define the terms: i) Screening Effect ii) Vander Waal's Radius [3]
iii) Isoelectronic Species [2]
b) Compare the size of Cl^- , Ar , & K^+ with reason. [2]
4. Justify the statement, "Oxidation and Reduction reaction occurs simultaneously." [2]
b) Define oxidation and reduction in terms of oxidation number. [1]
c) Find the oxidation number of:
i) S in $H_2S_2O_8$ ii) Ag in $K[Ag(CN)_2]$ [2]
5. a) Define autoredox reaction. [1]
b) Balance the following redox reaction either by oxidation number method or by ion electron method. [3]
 $Cl_2 + NaOH \longrightarrow NaCl + NaClO_3 + H_2O$
c) Identify oxidizing agent and reducing agent. [1]
6. a) State and explain Graham's law of diffusion. [2]
b) A gaseous hydrocarbon having the molecular formula C_nH_{2n+2} diffuses twice as quickly as sulphur dioxide. Determine the value of 'n' and molecular formula of hydrocarbon. [3]
7. a) State and explain Dalton's law of partial pressure. [2]

- b) 500 cc of oxygen under 680 mm pressure and 250 cc of nitrogen at 720 mm pressure were put together in a litre flask keeping the temperature constant. Find the final pressure of the mixture. [3]
8. a) Differentiate between oxidation number and valency. [2]
- b) Balance the following redox reaction either by oxidation number method or by ion electron method. [3]
- $$KMnO_4 + KCl + H_2SO_4 \longrightarrow MnSO_4 + K_2SO_4 + H_2O + Cl_2$$

GROUP - 'C' [3x8=24]

LONG ANSWER QUESTIONS

9. a) Derive Ideal gas equation. [2]
- b) Find the value of R in CGS unit. [2]
- c) Define absolute zero temperature. [1]
- d) An evacuated vessel weighs 50 gm when empty, 148 gm when filled with a liquid of density of 0.98 gm/cc and 50.5 gm when filled with an ideal gas at 760 mm Hg and at 300 K. Determine the molecular mass of gas. [3]
10. a) Explain with diagram and reaction involved for the preparation of H_2S using Kipp's apparatus. [3]
- b) Write the reaction to show that H_2S acts as precipitating agent. [1]
- c) Write the principle of manufacture of sulphuric acid by contact process. Also draw the flow sheet diagram for contact process. [2+2]
11. a) Write the principle for balancing redox reaction by ion-electron method. [1]
- b) Balance the following redox reaction either by oxidation number method or by ion electron method.
- $$CrO_4^{2-} + HSnO_2^{--} \longrightarrow CrO_2^- + HSnO_3^- \text{ (in alkaline medium)} \quad [4]$$
- c) A vessel contains 12 gm of an ideal gas at $t^\circ C$ temperature and 1 atm pressure. When the temperature is increased by $10^\circ C$ at the same volume, the pressure is increased by 10%. Calculate the volume of gas and initial temperature. [Molecular mass of gas = 120 amu]. [3]

FIRST TERM EXAMINATION 2082

Class: XI 'Science'

SET-B

Time: 25 Min

Subject: Com. Mathematics (0071)

Write the best option among the four options on your answer sheet.

GROUP 'A' MCQ [11×1=11]

Re-write the correct option of each question in your answer sheet.

1. If $x, y > 0$ and $\frac{1}{x} < \frac{1}{y}$, which one of the following is true?

A) $x > y$ B) $-\frac{1}{x} > -\frac{1}{y}$ C) $x < -y$ D) $x < y$

2. Which one of the following is equivalent to $\sim(p \wedge \sim q)$?

A) $\sim(p \wedge q)$ B) $(p \vee q)$ C) $(\sim p \vee q)$ D) $(\sim p \wedge q)$

3. What is the value of the $\lim_{x \rightarrow 1} \frac{\sin \pi x}{1-x^2}$?

A) $\frac{\pi}{2}$ B) π C) $\frac{2}{\pi}$ D) Does not exist

4. If the function $f(x) = \begin{cases} \frac{\sin^2 ax}{x^2} & \text{for } x \neq 0 \\ 16 & \text{for } x = 0 \end{cases}$ is continuous at $x = 0$, then what is the value of a ?

A) 16 B) 8 C) 4 D) 2

5. What is the value of $\lim_{t \rightarrow \infty} \frac{\sqrt{3t^2+1}}{3t-7}$?

A) $\frac{7}{\sqrt{3}}$ B) $\sqrt{3}$ C) $\frac{1}{\sqrt{3}}$ D) $\frac{-1}{\sqrt{3}}$

6. What is the value of $\lim_{y \rightarrow \pi} \frac{\sin y}{2+\cos y}$?

A) 0 B) 1 C) π D) 2

7. If $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ and $\mathbb{R}$ have their usual meaning, then which one of the following statements is not true?

A) $\mathbb{Z} \cap \mathbb{Q} = \mathbb{Z}$ B) $\mathbb{Z} \subset \mathbb{Q}$ C) $\mathbb{R} \subset \mathbb{Z}$ D) $\mathbb{Q} \cup \mathbb{R} = \mathbb{R}$

8. What is the value of $\lim_{x \rightarrow 0^+} (\frac{1}{x} - \frac{1}{|x|})$?

A) -1

B) 0

C) ∞

D) 1

9. What is the value of $\frac{\Delta y}{\Delta x}$, if $y = 2 - 3x$?

A) 2

B) -3

C) 5

D) 3

10. If $A = \{1, 2, 3\}$, what is the value of $n[P(A)]$?

A) 4

B) 15

C) 16

D) 8

11. What does $x \notin (B - A)$ mean?

A) $x \in A$ and $x \notin B$

C) $x \in B$ and $x \notin A$

B) $x \in A$ or $x \notin B$

D) $x \notin A$ or $x \in B$

Class: XI 'Science'

Time: 3 hrs.

Subject: Com. Mathematics (0071)

*Candidates are required to give their answers in their own words as far as practicable.
 The figures in the margin indicate full marks.*

Attempt all the questions.

GROUP 'B' [8×5=40]

- | | | |
|--------|--|-------|
| 12. a) | Define symmetric difference of two sets. | [1] |
| b) | For what value of x , $\sqrt{x}$ is real a number? | [1] |
| c) | What are conjugates of conditional $p \Rightarrow q$? | [1] |
| d) | Define a closed interval. | [1] |
| e) | Define the contradiction of a statement. | [1] |
| | | |
| 13. a) | If $P \subseteq Q$, then show that $\bar{Q} \subseteq \bar{P}$. | [2] |
| b) | For any three sets X, Y and Z , show that
$(X \cup Y) - Z = (X - Z) \cup (Y - Z)$ | [3] |
| | | |
| 14. a) | Write the antecedent and the consequent of the statement
<i>"If 5 is a natural number then $\frac{1}{5}$ is a rational number."</i> | [2] |
| b) | Let p, q and r be any statements. Show that the following statement is true.
$(p \Rightarrow q) \wedge (q \Rightarrow r) \Rightarrow (p \Rightarrow r)$ | [3] |
| | | |
| 15. a) | Write any two properties of inequalities. | [2] |
| b) | Solve the inequality: $-3 < 7 - 2x \leq 5$.
Also, represent the solution set on the number line. | [2+1] |
| | | |
| 16. a) | For any two sets A and B , prove that
$\overline{(A \cap B)} = \bar{A} \cup \bar{B}$ | [2] |
| b) | If $A = [-1, 3]$ and $B = (0, 2)$ then find $A \cup B, A \cap B$ and $A - B$ | [3] |
| | | |
| 17. a) | Let $h(x)$ be a continuous function. What is the relationship between $\lim_{x \rightarrow p} h(x)$ and $h(p)$? | [1] |
| b) | Define derivative of $f(x)$ with respect to x . | [1] |
| c) | If $\lim_{x \rightarrow c} f(x)$ exists, what is the relationship between
$\lim_{x \rightarrow c^-} f(x)$ and $\lim_{x \rightarrow c^+} f(x)$? | [1] |
| d) | Define jump discontinuity of a function $f(x)$ at $x = a$. | [1] |
| e) | What is the value of $\lim_{x \rightarrow 0} \frac{e^{\sqrt{2}x} - 1}{x}$? | [1] |

18. a) Evaluate the limit, if it exists. $\lim_{x \rightarrow 16} \frac{4 - \sqrt{x}}{16x - x^2}$ [2]
 b) If $f(x) = \begin{cases} \frac{\sqrt{1+px} - \sqrt{1-px}}{x}, & -1 \leq x < 0 \\ \frac{2x+1}{x-2}, & 0 \leq x \leq 1 \end{cases}$ [3]
 is continuous at $x = 0$. Find the value of p .

19. a) Find the limit of the following function if it exists: [2]
 $f(x) = \begin{cases} 2-x & \text{for } x < 2 \\ 2+x & \text{for } x \geq 2 \end{cases}$ at $x = 2$.
 b) i) Evaluate: $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x}$ [2]
 ii) Is it possible to write $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = \lim_{x \rightarrow 0} x^2 \times \lim_{x \rightarrow 0} \sin \frac{1}{x}$? If not, give reason. [1]

GROUP 'C' (3×8=24)

20. a) For any two statements p and q , prove that $\sim(p \vee q) \equiv \sim p \wedge \sim q$. [2]
 b) Define a **Statement**. How does it differ from **sentence** in the context of logic? Explain. [2]
 c) What is the relationship between the **existence of limits** and the **continuity of a function** at a particular point? Explain. [4]

21. a) Let $f(x) = \begin{cases} \frac{\tan x - \cot x}{x - \frac{\pi}{4}}, & x \neq \frac{\pi}{4} \\ a, & x = \frac{\pi}{4} \end{cases}$ be continuous at $x = \frac{\pi}{4}$. Find [4]
 the value of a .
 b) Find the derivative of the function [4]
 $f(x) = \sqrt[3]{(2-3x)^2}$ by using definition of the derivative.

22. a) Evaluate: $\lim_{x \rightarrow y} \frac{\tan x - \tan y}{x - y}$ [2]
 b) Find from the first principles, the differential coefficient of the [3]
 function $\cot(b - ax)$.
 c) If A and B be any two non-empty sets, verify that [3]
 $A - B$ and $B - A$ are disjoint sets by choosing suitable sets A and B .



KATHMANDU MODEL SECONDARY SCHOOL

FIRST TERM EXAMINATION 2082

Class: XI 'Science'

SET 'B'

F.M.: 50

Time: 2 hrs.

Subject: Computer Science (4271)

P.M.: 20

Candidates are required to give their answer in their own words as far as practicable.

The figure in the margin indicated full marks:

Attempt all the questions.

GROUP 'A' [9×1=9]

MULTIPLE CHOICE QUESTIONS.

- The three main components of computer are
 - CPU, Memory, Tape
 - I/O, Printer, Mouse
 - Tape, I/O, Floppy disk
 - CPU, I/O, Memory
- Which type of computer is smaller than a mainframe but larger than a microcomputer, and used by medium-sized organizations?
 - Hybrid Computers
 - Super Computers
 - Mini Computers
 - Analog Computers
- Which is the major component of fourth generation computers?
 - Transistor
 - Microprocessor
 - Vacuum Tube
 - IC
- A speaker is an example of device.
 - processing
 - storage
 - input
 - output
- What is the binary equivalent of a decimal number 31?
 - 10101
 - 11111
 - 11010
 - 101101
- How many bits are there in 2.5 kilobytes?
 - 20,480 bits
 - 25,600 bits
 - 40,960 bits
 - 20,480 bytes
- The maximum number of bits used to represent hexadecimal number in binary is
 - 3
 - 4
 - 8
 - 16
- The _____ is a small input device used to select and display objects on a screen.
 - Ink
 - Magnet
 - Light pen
 - None of these
- Which of the following is not the font style?
 - Bold
 - Italics
 - Regular
 - Superscript

1024 bytes

KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2082

Part: XI Science

Time: 25 min.

Subject: Physics [1011]

SET B

Give the best option among the four options on your answer sheet.

Give the best alternative.

GROUP 'A' [11-1-11]

Multiple choice questions

1. Turpentine oil is flowing through a tube of length l' and radius r' . The pressure difference between the two ends of the tube is P . The viscosity of the oil is given by: $\eta = \frac{P}{4\pi l' r'^2} \frac{V}{\Delta t}$, where V is the velocity of oil at a distance x from the axis of the tube. The dimensions of η are:

a) $[M^0 L^0 T^0]$

b) $[M^1 L^2 T^{-1}]$

c) $[M^1 L^2 T^{-2}]$

d) $[M^1 L^{-1} T^{-1}]$

2. The expression $\left[\left(\frac{1}{\sqrt{3}} i - \frac{1}{\sqrt{3}} j + \frac{1}{\sqrt{3}} k \right) \right]$ is a:

a) Null vector

b) Unit vector

c) Vector of magnitude $\sqrt{2}$

d) Scalar

3. From the figure:

a) $\vec{A} + \vec{B} + \vec{C} + \vec{D} = \vec{0}$

b) $\vec{A} + \vec{B} + \vec{C} + \vec{D} + \vec{E} = \vec{0}$

c) $\vec{A} + \vec{B} + \vec{C} + \vec{D} - \vec{E} = \vec{0}$

d) $\vec{A} + \vec{B} + \vec{C} + \vec{D} - 2\vec{E} = \vec{0}$



4. What is the angle between $(\vec{P} + \vec{Q})$ and $(2\vec{P} + 2\vec{Q})$?

a) Zero

b) π

c) $\frac{\pi}{2}$

d) π

5. An aeroplane is moving in a circular path with a speed of 125 km/hr, what is the change in velocity in half revolution?

a. 500 Km/hr

b. 250 Km/hr

c. 125 Km/hr

d. Zero:

6. A certain vector in x-y plane has an x-component of 12 m and y-component of 8m. It is then, rotated in xy plane so that its y-component is halved. Then its new x component is approximately

a) 14m

b) 10.12m

c) 2.14 m

d) 13.93 m

7. The figure shows some of the electric field lines corresponding to an electric field. The figure suggests:

a) $E_A = E_C > E_B$

b) $E_A = E_C < E_B$

c) $E_A > E_B > E_C$

d) $E_A = E_B = E_C$



8. An isolated conducting sphere is given a negative charge. Its mass

a) gets increased

b) gets decreased

c) remains the same

d) mass is not involved during electrification

9. The magnitude of electric field strength E such that an electron placed in it would experience an electrical force equal to its weight, is given by:

a) mg

b) mg/e

c) mg/e^2

d) e/mg

10. A temperature degree on Kelvin Scale is same as:

a) a temperature degree on Celsius Scale

b) a temperature degree on Fahrenheit Scale

c) a temperature degree on Rankine Scale

d) $1/32$ th part of temperature degree on Fahrenheit Scale

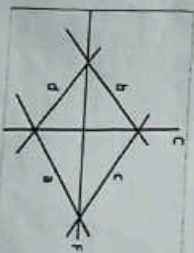
11. Which of the curve in the figure represents the relation between Celsius and Fahrenheit temperatures?

a) a

b) b

c) c

d) d



KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2082

Grade: XI Science

Time: 25min.

Give the best option among the four options on your answer sheet.

Subject: Physics [1011]
SET B

Choose the best alternative.

GROUP 'A' [11x1=11]

Multiple choice questions

1. Turpentine oil is flowing through a tube of length 'l' and radius 'r'. The pressure difference between the two ends of the tube is P. The viscosity of the oil is given by: $\eta = \frac{P}{4\pi l} (r^2 - r'^2)$, where 'V' is the velocity of oil at a distance x from the axis of the tube. The dimensions of η are:
a) $[M^0 L^2 T^{-1}]$ b) $[M^2 L^2 T^{-1}]$ c) $[M^1 L^2 T^{-2}]$ d) $[M^1 L^{-1} T^{-1}]$

2. The expression $\left[\left(\frac{1}{\sqrt{3}} \hat{i} - \frac{1}{\sqrt{3}} \hat{j} + \frac{1}{\sqrt{3}} \hat{k} \right) \right]$ is a:
a) Null vector b) Unit vector
c) Vector of magnitude $\sqrt{2}$ d) Scalar

3. From the figure:

- a) $\vec{A} + \vec{B} + \vec{C} + \vec{D} = \vec{0}$
- b) $\vec{A} + \vec{B} + \vec{C} + \vec{D} + \vec{E} = \vec{0}$
- c) $\vec{A} + \vec{B} + \vec{C} + \vec{D} - \vec{E} = \vec{0}$
- d) $\vec{A} + \vec{B} + \vec{C} + \vec{D} - 2\vec{E} = \vec{0}$

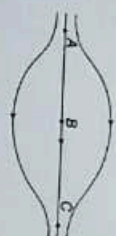


4. What is the angle between $(\vec{P} + \vec{Q})$ and $(2\vec{P} + 2\vec{Q})$?
a) Zero b) π c) $\frac{\pi}{2}$ d) π

5. An aeroplane is moving in a circular path with a speed of 125 Km/hr. What is the change in velocity in half revolution?
a. 500 Km/hr b. 250 Km/hr c. 125 Km/hr d. Zero.

6. A certain vector in xy plane has an x-component of 12 m and y-component of 8m. It is then rotated in xy plane so that its y-component is halved. Then its new x component is approximately
a) 14m b) 10.12m c) 2.14 m d) 13.93 m

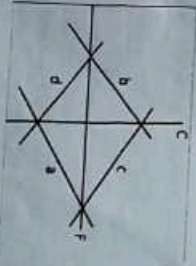
7. The figure shows some of the electric field lines corresponding to an electric field. The figure suggests:
a) $E_A = E_C > E_B$
b) $E_A = E_C < E_B$
c) $E_A > E_B > E_C$
d) $E_A = E_B = E_C$



8. An isolated conducting sphere is given a negative charge. Its mass
a) gets increased b) gets decreased
c) remains the same d) mass is not involved during electrification
9. The magnitude of electric field strength E such that an electron placed in it would experience an electrical force equal to its weight, is given by:
a) mg b) mg/e c) mg/e^2 d) e/mg

10. A temperature degree on Kelvin Scale is same as:
a) a temperature degree on Celsius Scale
b) a temperature degree on Fahrenheit Scale
c) a temperature degree on Rankine Scale
d) 1/32th part of temperature degree on Fahrenheit Scale

11. Which of the curve in the figure represents the relation between Celsius and Fahrenheit temperatures?
a) a b) b c) c d) d





KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2082

Class: XI Science

Time: 25min.

Subject: Physics [1011]

SET A

Write the best option among the four options on your answer sheet.

Choose the best alternative.

Multiple choice questions

GROUP 'A' [1×1=11]

1. Turpentine oil is flowing through a tube of length l' and radius r' . The pressure difference between the two ends of the tube is P . The viscosity of the oil is given by: $\eta = \frac{P}{4\pi l' r'^2} x^2$, where x is the velocity of oil at a distance x from the axis of the tube. The dimensions of η are:
a) $[M^2 L^2 T^{-1}]$ b) $[M^1 L^2 T^{-1}]$ c) $[M^1 L^2 T^{-2}]$ d) $[M^1 L^{-1} T^{-1}]$

2. The expression $\left[\frac{1}{\sqrt{2}}\hat{i} - \frac{1}{\sqrt{2}}\hat{j}\right]$ is a
a) Unit vector b) Null vector
c) Vector of magnitude $\sqrt{2}$ d) Scalar

3. From the figure:

- a) $\vec{A} + \vec{B} = \vec{C}$
b) $\vec{B} + \vec{C} = \vec{A}$
c) $\vec{C} + \vec{A} = \vec{B}$
d) $\vec{A} + \vec{B} + \vec{C} = \vec{0}$

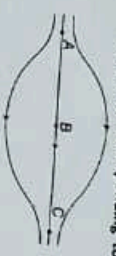


4. What is the angle between $(\vec{P} + \vec{Q})$ and $(\vec{P} - \vec{Q})$?
a) Zero b) π c) $\frac{\pi}{2}$ d) π

5. A force of 120 N and a force of 20N acts simultaneously at a point may produce a resultant force of:
a) 80 N b) 140 N c) 160 N d) 0 N

6. A certain vector in xy plane has an x -component of 12 m and y -component of 8m. It is then, rotated in xy plane so that its x -component is halved. Then its new y component is approximately
a) 14m b) 10.12m c) 2.14m d) 13.11m

7. The figure shows some of the electric field lines corresponding to an electric field. The figure suggests:
a) $E_A > E_B > E_C$
b) $E_A = E_B = E_C$
c) $E_A = E_C > E_B$
d) $E_A = E_C < E_B$

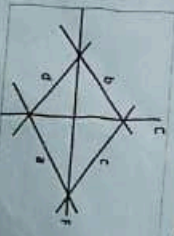


8. An isolated conducting sphere is given a positive charge. Its mass
a) gets increased b) gets decreased
c) remains the same d) mass is not involved during electrification

9. The magnitude of electric field strength E such that an electron placed in it would experience an electrical force equal to its weight, is given by:
a) mg b) mg/e c) mg/e^2 d) e/mg

10. Two thermometers are constructed in such a way that one has a spherical bulb and the other has a elongated cylindrical bulb. which one will respond more quickly to temperature changes?
a) with cylindrical bulb b) with spherical bulb
c) both respond similarly d) any one may respond quickly depending upon the value of temperature

11. Which of the curve in the figure represents the relation between Celsius and Fahrenheit temperatures?



- a) a
b) b
c) c
d) d

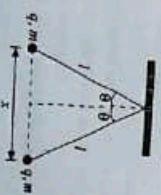
- b) State and explain the principle of superposition of Coulomb's force with suitable diagram. [2]
- c) A polythene piece rubbed with wool is found to have a $-ve$ charge of $0.64 \mu C$. (i) estimate the number of electrons transferred from which to which? (ii) is there a transfer of mass from wool to polythene? If yes, find the amount of mass transferred. [2]
18. a) Find the dimensions of Planck's constant. If its value in cgs system is $6.62 \times 10^{-27} \text{ erg-sec}$, what will be its value in MKS system? [2]
- b) Assuming that the mass m of the largest stone that can be moved by a flowing river depends upon velocity v , the density ρ and acceleration due to gravity g , show that mass m varies directly as the sixth power of the velocity of the flow. [3]

Long answer questions:

GROUP 'C' [3×8=24]

20. a) What is meant by resolution of a vector? Why resolving a vector is important in physics? Can the walking of a man be an example of resolution of vectors? Explain. [3]
- b) Can the magnitude of the rectangular component of a vector be greater than the magnitude of that vector. Explain. [1]
- c) A man rows a boat with a speed of 18 km h^{-1} in the north-west direction. The shoreline makes an angle of 15° south of west. Find the components of velocity of the boat along the shoreline and perpendicular to the shoreline. [2]
- d) An athlete runs 2 km , east and then 3 km , 40° south of east. Find the magnitude and direction of total displacement. [2]
21. a) What do you mean by absolute zero and absolute scale of temperature? [2]
- b) A student claimed that the thermometers register their own temperature therefore, they are useless. How will you respond? [2]

- c) Mercury boils at 357°C . However, mercury thermometers are made such that they can measure temperature up to 500°C . Explain how this is possible. [1]
- d) Suppose you come across old scientific notes that describe a temperature scale called Z on which the boiling point of water is 65.0°Z and the freezing point is -14.0°Z . To what temperature on Fahrenheit scale would a temperature of $T = -98.0^\circ\text{Z}$ correspond? Assume that Z scale is linear, that is, size of a Z degree is the same everywhere on the Z-scale. [1]
22. a) Explain modern theory of electrification? [3]
- b) Define 1 Coulomb of charge on the basis of Coulomb's law. [1]
- c) When a glass rod is rubbed with a silk cloth, charges appear on both. A similar phenomenon is observed with many other pairs of bodies. Explain how this observation is consistent with the law of conservation of charge. [1]
- d) Two small spheres each of mass m and positive charge q , hang from light threads of lengths l . Each thread makes an angle of θ with the vertical as shown in the figure. [2]



- (i) On the diagram draw and label all the forces on spheres. [1]
- (ii) Develop an expression for the charge q in terms of m , l , θ and g and the Coulomb's law constant. [3]

"Best of Luck"

Class: XI Science

Time: 3hrs.

Subject: Physics '1011'

F.M.: 75
P.M.: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions:

GROUP 'B' [8x5=40]

Short answer questions:

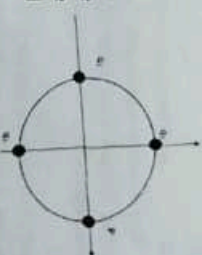
12. a) What are dimensional and non-dimensional variable? Give examples of each. [2]
- b) Check the correctness of the physical relation $Y = \frac{Mg}{\pi r^2 l}$ where Y = Young's modulus Mg is weight applied on wire, l is length of wire, r is increase in length of wire, and r is radius of wire. [2]
- c) How many times the units of energy be increased if each of the units of mass, length and time is increased by two times. [1]
13. a) Two people are pulling a heavy cart from the same point but in slightly different directions. One pulls with a force of magnitude A , the other with force B , and the angle between their ropes is θ . The cart doesn't turn—it moves straight in a direction that results from both pulls happening simultaneously. Using the rule that combines two vectors acting from the same point at an angle, derive an expression for the total force acting on the cart (its magnitude and direction). [3]
- b) The two forces are given by: $\vec{F}_1 = 500 \text{ N, east}$ and $\vec{F}_2 = 600 \text{ N, north}$. Draw a vector diagram to subtract $\vec{F}_2$ from $\vec{F}_1$. Find the magnitude and direction. [2]
14. a) A scalar may be a pure number or a physical quantity. What is the effect of multiplication of a vector by a scalar? [2]
- b) Give the geometrical interpretation of dot product of two vectors. [1]

- c) If $\vec{A} = 4\hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{B} = -2\hat{i} - 5\hat{j} + 7\hat{k}$, find the angle between $\vec{A}$ and $\vec{B}$. [2]

15. a) The expression of electric field E due to a point charge at any point near to it is defined by $\vec{E} = \lim_{q \rightarrow 0} \frac{\vec{F}}{q}$ where q is the test charge and F is the force acting on it. What is the significance of $q \rightarrow 0$ in this expression? [1]

- b) Why two electric field lines cannot intersect with each other? [1]

- c) Consider three equal charges Q_1, Q_2 , and $Q_3 = q = +1\mu\text{C}$ located at three different points on a circle of radius 1m , as shown in the figure. Calculate the total electric field at point P due to this distribution of charges. [3]



16. a) What are induced and inducing charges. How will you charge a conductor positively by the method of induction. Explain with clear diagram. [1+2]
- b) A small ball of paper has a mass of $9 \times 10^{-5} \text{ kg}$ and carries a charge of $5 \mu\text{C}$. When it is held over another charged ball of paper at a distance of 2 cm above it, the two balls stay in equilibrium, what is the charge on the second ball? [2]
17. a) What are electric field lines? [1]
- b) If one field line represents $1.0 \times 10^{-9} \text{ C}$ of charge, determine the number of field lines drawn for a charge of $2 \mu\text{C}$. [1]
- c) Two-point charges $+2 \mu\text{C}$ and $+8 \mu\text{C}$ are placed 10 cm apart in air. Sketch qualitatively the electric field lines, indicate the region and calculate the position where the resultant field will be zero. [3]
18. a) Write the conditions for validity of Coulomb's law of force between charges. [1]

Determine the magnitude of the electrostatic force on the electron at this location and state its direction. If the electron is released from rest at this location midway between the plates, determine its speed just before striking one of the plates. Assume that the gravitational effects are negligible.

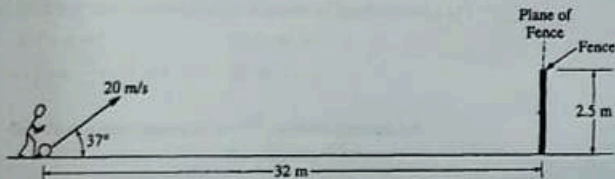
[1+2]

19. a) Derive a relation that shows the variation of density of a solid with temperature. A liquid has a density of 1000 Kg m^{-3} at 0°C . If its coefficient of linear expansion is $6.67 \times 10^{-5} / ^\circ\text{C}$, What will be its density when it heated to 50°C ? [2+1]
- b) A steel wire of diameter 2 mm is stretched between two rigid clamps with an initial tension of 100 N at a temperature of 40°C . If the temperature of the wire decreases to 10°C , calculate the new tension in the wire. (Given, $Y_{\text{steel}} = 2 \times 10^{11} \text{ N/m}^2$ and $\alpha_{\text{steel}} = 1.2 \times 10^{-5} / ^\circ\text{C}$) [2]

GROUP 'C' [3×8=24]

Long answer questions.

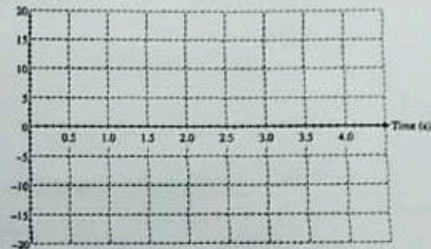
20. a) What is projectile motion. Show that with proper sketch the path followed by a horizontal projectile is parabolic in nature. [3]
- b)



A ball of mass 0.5 kilogram, initially at rest, is kicked as shown in figure.

- i) Determine the time it takes for the ball to reach the plane of the fence. [1]
- ii) Will the ball hit the fence? If so, how far below the top of the fence will it hit? If not, how far above the top of the fence will it pass? [1]
- iii) What will be the velocity when the ball is at the plane of the fence? [1]

- iv. On the axes below, sketch the horizontal and vertical components of the velocity of the ball as functions of time until the ball reaches the plane of the fence. [2]



21. a) State and explain Newton's law of cooling with its limitations. [3]
- b) In an experiment to determine the specific latent heat of vaporization of steam L_f , the following measurements were taken: mass of calorimeter + stirrer = x kg, mass of water = y kg, mass of steam condensed = z kg, initial temperature of water = $\theta_1^\circ\text{C}$, specific heat capacities of calorimeter and water are s_1 and s_2 respectively. Express L_f in terms of above data. [2]
- c) What are the difference between water equivalent and thermal capacity of a body? [1]
- d) From what height a block of ice is dropped in order that it may completely melt. It is assumed that 20% of energy of fall is retained by ice? ($L_f = 80 \text{ cal/gm}$) [2]
22. a) Define electric flux and find its dimensional formula. [2]
- b) State and prove Gauss theorem of Electrostatics. Write some properties of Gaussian surface. [1+1+1]
- c) A hollow spherical conductor of radius 12 cm has a uniform surface charge density of 13.26 nC/m^2 . Calculate the electric field strength at a distance (i) 6 cm and (ii) 4 m from the centre of the sphere. [1+2]

"Best of Luck"

Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL
SECOND TERM EXAMINATION 2082

Class: XI 'Science'

SET-B

Time: 25min.

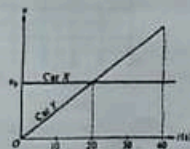
Subject: Physics [1011]

Tick the best option among the four options. Apply tick on two options will be automatically denied and mark will not be counted.

Choose the best alternative.

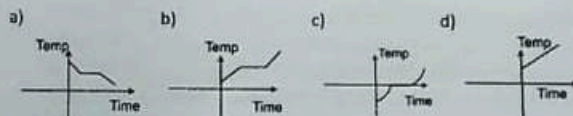
GROUP 'A' [11×1=11]

- In the expression, $\frac{1}{2} \epsilon_0 E^2$ (where ϵ_0 is permittivity and E is electric field) has same dimensions as that of:
 (a) Surface Tension (b) Force (c) Power (d) Pressure
- Given that $\vec{P} + \vec{Q} + \vec{R} = 0$, and $|\vec{P}| = 3$, $|\vec{Q}| = 5$ and $|\vec{R}| = 7$, what is the angle between $\vec{P}$ and $\vec{Q}$?
 (a) 120° (b) 60° (c) 45° (d) 30°
- A ball initially moves horizontally with velocity v_i as shown. It is then struck by a stick. After leaving the stick, the ball moves vertically with a velocity v_f , which is smaller in magnitude than v_i . Which of the following vectors best represents the direction of the average force that the stick exerts on the ball?
 (a)  (b)  (c)  (d) 
- At time $t = 0$, car X travelling with speed v_0 passes car Y which is just starting to move. Both cars then travel on two parallel lines of the same straight road. The graph of speed v versus time t for both cars is shown. Which of the following is true at time $t = 20$ seconds?
 (a) Car Y is behind car X
 (b) Car Y is passing car X
 (c) Car Y is in front of car X
 (d) Car X is accelerating faster than Car Y



a laboratory thermometer, because:

- It is marked in Fahrenheit not in Celsius
 - a long column in inconvenient
 - Only a limited range of temperature is required.
 - a long column of mercury cannot be introduced in the glass tube.
- Two copper rods of equal length but unequal diameter are heated through the same range of temperature. Its increase in length will be:
 (a) more for thinner rod (b) more for thicker rod
 (c) same for both the rods (d) considerable for thinner rod
 - A concave mirror with a radius of curvature of 1.0 m is used to collect light from a distant star. The distance between the mirror and the image of the star is most nearly
 (a) 0.25 m (b) 0.50 m (c) 0.75 m (d) 1.0 m
 - Liquid oxygen at 50 K is heated to 300 K at constant pressure of 1 atm. The rate of heating is constant. Which one of the following graphs represents the variation of temperature with time?



- one gram hydrogen atom has 6×10^{23} atoms. Imagine that all the nuclei are put at the north pole of the earth and the electrons are at the south pole of the earth (radius 6400 km). The force between the charges is:
 (a) $1.0 \times 10^6 N$ (b) $5.0 \times 10^5 N$
 (c) $2.5 \times 10^5 N$ (d) $2.0 \times 10^5 N$
- A hollow cylinder has a charge q coulomb within it. If ϕ is the electric flux in random unit associated with the curved surface B, the flux linked with the plane surface A in the same units will be:
 (a) $\frac{q}{2\epsilon_0}$ (b) $\frac{\phi}{3}$ (c) $\frac{q}{\epsilon_0} - \phi$ (d) $\frac{1}{2} \left(\frac{q}{\epsilon_0} - \phi \right)$



- If electric field is uniform, then the electric lines of force are:

(a) divergent (b) convergent (c) circular (d) parallel

Class: XI 'Science'

Time: 3hrs.

Subject: Physics '1011'

F.M.: 75

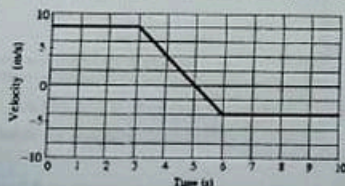
P.M.: 30

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

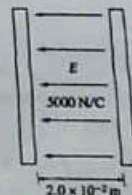
Attempt all the questions:

GROUP 'B' [8×5=40]

12. a) The volume per second of a liquid flowing through a horizontal pipe of length l is given by $\frac{kpa^x}{\eta}$, where k is constant, p is the excess pressure (force per unit area), a is the radius of the pipe and η is a frictional quantity of dimensions $[ML^{-1}T^{-1}]$. By dimensionally find the value of x . [2]
- b) A swimmer can swim in still water with a speed of 2 m/s. He intends to cross a river 200 m wide. The river flow is non-uniform: in the first 100 m, the current speed is 1 m/s, but in the second 100 m the current increases to 1.5 m/s. Find, (i) the minimum time of crossing the river (ii) total time taken to cross the river if the swimmer reaches a point exactly opposite to his starting point [1+2]
13. a) What is instantaneous velocity? How will you find its value from the position-time graph of a moving object? Explain. [2]
- b) The velocity vs. time graph for the motion of a car on a straight track is shown in the diagram. The thick line represents the velocity. Assume that the car starts at the origin $x = 0$.
i) At which time, the car is at greatest distance from the origin? [1]
ii) What is the average velocity of the car for the 10 second interval? [2]
14. a) Derive the relation between superficial and cubical expansivity of the material. [3]



- b) Pendulum of a clock gives correct time at 20°C . How many seconds a day will it gain or lose when the temperature falls to 5°C ? ($\alpha_{\text{pendulum}} = 0.0002/^\circ\text{C}$) [2]
15. a) What is meant by relative velocity? Write the general expression to obtain the relative velocity with appropriate diagram. [2]
- b) To a man walking due east at the rate of 2 kmh^{-1} , rain appears to fall vertically. When he increases his speed to 4 kmh^{-1} , it appears to meet him at an angle of 45° . Find the real direction and speed of the rain. [3]
16. a) Derive a relation for Radius of curvature, object distance and image distance for a concave mirror when virtual image is formed? [3]
- b) An object 2cm in height is placed at 30 cm from a convex mirror of radius of curvature 60 cm. Find the position and size of image. [2]
17. a) State the principle of conservation of linear momentum. [1]
- b) Prove the principle for a system of two interacting bodies using Newton's third law of motion. [2]
- c) A ball A of mass 0.1 kg moving with a velocity of 6 m/s collides directly with a ball B of mass 0.2 kg at rest. Calculate their common velocity if both balls move off together. If A had rebounded with a velocity of 2 m/s in the opposite direction after collision, what would be the new velocity of B? [2]
18. a) What are electric field lines? How does the density of electric field lines change when they enter a dielectric medium (like glass or water) from a vacuum? [2]
- b) Two parallel conducting plates, each of area 0.30 m^2 are separated by a distance of $2.0 \times 10^{-2}\text{ m}$ of air. One plate has charge $+Q$ and another plate has charge $-Q$. An electric field of 5000 N/C is directed to the left in the space between the plates, as shown in the diagram.
i) Indicate by redrawing diagram which plate is +ve and which plate is -ve.
ii) An electron is initially located at a point midway between the plates.



KATHMANDU MODEL SECONDARY SCHOOL

SECOND TERM EXAMINATION 2082

Class: XI 'Science'

SET-A

Time: 25min.

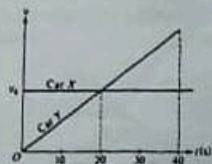
Subject: Physics [1011]

Tick the best option among the four options. Apply tick on two options will be automatically denied and mark will not be counted.

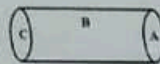
Choose the best alternative.

GROUP 'A' [11×1=11]

- In the expression, $v = At^2 + \frac{B}{C+t}$, where v is velocity and t is time, what is the dimensional formula for the product ABC ?
 (a) $[ML^2T^{-2}]$ (b) $[L^2T^{-2}]$
 (c) $[L^2T^{-3}]$ (d) $[M^0L^2T^0]$
- If the two vectors $\vec{P} = \hat{i} + 2m\hat{j} + m\hat{k}$ and $\vec{Q} = 4\hat{i} - 2\hat{j} + m\hat{k}$ are perpendicular to each other, what is the value of m ?
 (a) 1 or 2 (b) -1 or 2 (c) 2 or 2 (d) 1 or -2
- A ball initially moves horizontally with velocity v_0 as shown. It is then struck by a stick. After leaving the stick, the ball moves vertically with a velocity v_1 , which is greater in magnitude than v_0 . Which of the following vectors best represents the direction of the average force that the stick exerts on the ball?
 (a) (b) (c) (d)
- At time $t = 0$, car X travelling with speed v_0 passes car Y which is just starting to move. Both cars then travel on two parallel lines of the same straight road. The graph of speed v versus time t for both cars is shown. From time $t = 0$ to time $t = 40$ seconds, the areas under both curves are equal. Therefore, which of the following is true at time $t = 40$ seconds?
 (a) Car Y is behind car X. (b) Car Y is passing car X.



- When a liquid is heated in a glass vessel, the level of liquid first falls slightly before it begins to rise. This initial fall is due to:
 (a) The evaporation of liquid (b) Contraction of liquid
 (c) The anomalous expansion of liquid (d) The expansion of glass vessel
- A bimetallic strip made of brass and iron is heated. Brass has a higher coefficient of linear expansion than iron. In which direction will the strip bend?
 (a) Towards the iron side (b) It will not bend, only increase in length
 (c) towards the brass side (d) it will twist into spiral
- Liquid oxygen at 50 K is heated to 300 K at constant pressure of 1 atm. The rate of heating is constant. Which one of the following graphs represents the variation of temperature with time?
 (a) (b) (c) (d)
- A ray of light passing through the centre of curvature of a concave mirror is reflected back along the same path. Why?
 (a) The mirror is perfectly flat at that point (b) The angle of incidence is 0°
 (c) The ray is parallel to principal axis (d) The angle of incidence is 90°
- If the radius of a Gaussian sphere enclosing a charge is halved, how does the total electric flux through the surface change?
 (a) It becomes four times larger (b) It remains the same
 (c) It becomes half (d) It becomes zero
- A hollow cylinder has a charge q coulomb within it. If ϕ is the electric flux in random unit associated with the curved surface B, the flux linked with the plane surface C in the same units will be:
 (a) $\frac{q}{2\epsilon_0}$ (b) $\frac{\phi}{3}$ (c) $\frac{q}{\epsilon_0} - \phi$ (d) $\frac{1}{2}(\frac{q}{\epsilon_0} - \phi)$
- Which of the following values of dielectric constant K is impossible for any material?
 (a) 80 (b) 0.5 (c) 1.0 (d) infinity





Class: XI 'Science'

SET-A

Time: 25min.

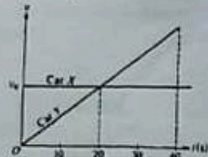
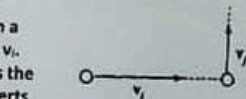
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Choose the best alternative.

GROUP 'A' [11×1=11]

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(a) 1 or 2 (b) -1 or 2 (c) 2 or 2 (d) 1 or -2
- A ball initially moves horizontally with velocity v_i as shown. It is then struck by a stick. After leaving the stick, the ball moves vertically with a velocity v_f , which is greater in magnitude than v_i . Which of the following vectors best represents the direction of the average force that the stick exerts on the ball?
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- Which of the following values of dielectric constant K is impossible for any material?
(a) 80 (b) 0.5 (c) 1.0 (d) infinity

Advancing Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL
 SECOND TERM EXAMINATION 2082

Class: XI 'Science'

Time: 3hrs.

Subject: Chemistry [3011]

F.M.: 75

P.M.: 30

Candidates are required to give their answers in their own words as far as practicable.
 The figures in the margin indicate full marks.

Attempt all the questions.

GROUP - 'B' [8x5=40]

SHORT ANSWER QUESTIONS.

1. a) Show with example that all reactions are not redox reaction. [1]
 b) Balance the following redox reaction either by oxidation number method or by ion electron method. [4]

$$C_{12}H_{22}O_{11} + HNO_3 \longrightarrow H_2C_2O_4 + NO_2 + H_2O$$
2. a) State Faraday's First Law of Electrolysis. [1]
 b) Define 1 Faraday. [1]
 c) Iron slab having dimension 50 cm x 30 cm x 20 cm was electroplated all around the surface with a uniform thickness of copper by passing 5 amperes of current for 2 hours in $CuSO_4$ solution. Find the thickness of Cu deposited. Given, the specific gravity of Cu is 8.9 g/cc and its atomic mass is 63.5 amu. [3]
3. a) Write the limitations of Rutherford's atomic model. [1]
 b) How did Bohr interpreted origin of hydrogen spectra on the basis of his atomic model? [4]
4. a) 390 g of SO_2 were treated with 170g O_2 and 120 g water to form H_2SO_4 with the help of following reaction.

$$2SO_2 + O_2 + 2H_2O \longrightarrow 2H_2SO_4$$

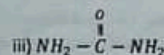
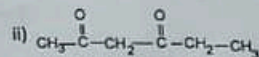
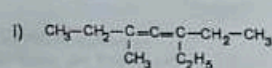
 Find out the limiting reactant and weight in gram and moles of reactants left behind. [3]
 b) What weight of 60% pure H_2SO_4 is required to decompose 25g of Chalk ($CaCO_3$)? [2]
5. a) Differentiate between real gas and ideal gas. Write the condition in which real gas shown ideal behavior. [1+1]
 b) A mixture of ozone and oxygen containing 20% by volume of ozone diffused through a porous plug in 172 seconds, while the same volume of pure oxygen took 164 seconds to diffuse through the same plug. Calculate the relative density and molecular mass of ozone. [3]

6. a) Define electro negativity. [1]
 b) Discuss any four factors that affect electronegativity. [2]
 c) Why is electron affinity of chlorine higher than that of fluorine? [2]
7. a) Write the principle for manufacture of sulphuric acid by contact process. [2]
 b) Give one reaction to show acidic nature of H_2SO_4 . [1]
 c) Compare the bleaching action of SO_2 with Cl_2 . [2]
8. a) Define octane number. A fuel has an octane number 75. What does it mean? [1+1]
 b) Write the reaction for detection of nitrogen as hetero element. [2]
 c) Sodium extract solution must be boiled with HNO_3 before test for halogen, why? [1]

GROUP - 'C' [3x8=24]

LONG ANSWER QUESTION

9. a) State Heisenberg's uncertainty principle. [1]
 b) Why is Bohr's atomic model against the concept of Heisenberg's uncertainty principle? [2]
 c) Write the values of all four quantum numbers for 11th electron of sodium. [2]
 d) Write the differences between orbit and orbital. [3]
10. a) State Avogadro's hypothesis and hence derive the relationship between molecular weight and vapour density. [1+2]
 b) State the law of reciprocal proportion with example. [2]
 c) A hydrocarbon C_xH_y has mass ratio between hydrogen and carbon 1:10.5. One Liter of the hydrocarbon at 127°C and 1 atm pressure weights 2.8g. Find the molecular formula of the hydrocarbon. [3]
11. a) Write the IUPAC name of: [1+1+1]



- b) Define homologous series and write its four characteristics. [1+2]
- c) Define Heterocyclic compounds with example. [1]
- d) Write a chemical reaction to show reducing nature of H_2S . [1]

Best of Luck

GROUP - 'A' [11×1 = 11]

Multiple Choice Questions

- The set of quantum numbers not applicable for an electron in an atom is
 a) $n=1, l=1, m=1, s=+\frac{1}{2}$ b) $n=1, l=0, m=0, s=+\frac{1}{2}$
 c) $n=1, l=0, m=0, s=-\frac{1}{2}$ d) $n=2, l=0, m=0, s=+\frac{1}{2}$
- If uncertainty in position of an electron is zero, the uncertainty in its momentum would be
 a) Zero b) $> \frac{h}{4\pi}$ c) $< \frac{h}{4\pi}$ d) infinite
- A smuggler could not carry gold by depositing iron on the gold surface since
 a) gold is denser
 b) iron rusts
 c) reduction potential of gold is higher than iron
 d) reduction potential of gold is lower than iron
- $xMnO_4^- + yH_2O_2 \rightarrow 2Mn^{2+} + 5H_2O + 9O_2 + ze^-$ In this reaction, value of x, y and z are
 a) 2, 5, 6 b) 5, 2, 9 c) 3, 5, 5 d) 2, 6, 6
- In Lassaigne's test for detection of sulphur, a violet colour is obtained when sodium nitroprusside is added to sodium extract. It is due to the formation of

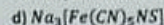
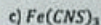
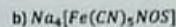
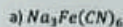
- a) $Na_3Fe(CN)_6$ b) $Na_4[Fe(CN)_5NOS]$
 c) $Fe(CNS)_3$ d) $Na_3[Fe(CN)_5NS]$

- Different proportions of oxygen in the various oxides of nitrogen proves the law of
 a) equivalent proportion b) multiple proportion
 c) constant proportion d) conservation of mass
- The pair of gases containing same number of molecules is
 a) 22 g of CO_2 and 72 g of N_2 b) 11 g of CO_2 and 28 gm of N_2
 c) 44 gm of CO_2 and 7 g of N_2 d) 11 g of CO_2 and 7 g of N_2
- If the four tubes of a car are filled to the same pressure with N_2, O_2, H_2 and Ne separately, then which one will be filled first?
 a) N_2 b) O_2 c) H_2 d) Ne
- Correct order for second ionization energy is
 a) $Be < C < B < Li$ b) $Be < Li < B < C$
 c) $Li < Be < B < C$ d) $Be > Li < B < C$
- The most stable allotrope of sulphur is
 a) β - sulphur b) α - sulphur
 c) γ - sulphur d) δ - sulphur
- The IUPAC name of the compound $CH_3-\overset{\overset{H}{|}}{\underset{\underset{C_2H_5}{|}}{C}}-\overset{\overset{C_4H_9}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_3$ is
 a) 2 - butyl - 2 - methyl - 3 - ethylbutane
 b) 2 - ethyl - 3, 3 - dimethylheptane
 c) 3, 4, 4 - trimethylheptane
 d) 3, 4, 4 - trimethyl octane

GROUP - 'A' [11×1 = 11]

Multiple Choice Questions

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 a) $n=1, l=1, m=1, s=+\frac{1}{2}$ b) $n=1, l=0, m=0, s=+\frac{1}{2}$
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- $x\text{MnO}_4^- + y\text{H}_2\text{O}_2 \rightarrow 2\text{Mn}^{2+} + 5\text{H}_2\text{O} + 9\text{O}_2 + ze^-$ In this reaction, value of x , y and z are
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 a) 2 - butyl - 2 - methyl - 3 - ethylbutane
 b) 2 - ethyl - 3, 3 - dimethylheptane
 c) 3, 4, 4 - trimethylheptane
 d) 3, 4, 4 - trimethyl octane

GROUP - 'A' [11×1 = 11]

Multiple Choice Questions

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KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2082

Just XI Science
Time: 25 Min
Subject: Com. Mathematics (0071)
SET-B
Give the best option among the four options on your answer sheet.

GROUP A: MCQ (1×11=11)

Write the correct option of each question in your answer sheet.

1. If $x, y > 0$ and $\frac{1}{x} < \frac{1}{y}$, which one of the following is true?
☒ A) $x > y$ ☐ B) $-\frac{1}{x} > -\frac{1}{y}$ ☐ C) $x < -y$ ☐ D) $x < y$
2. Which one of the following is equivalent to $\sim(p \wedge \sim q)$?
☐ A) $\sim(p \wedge q)$ ☐ B) $(p \vee q)$ ☒ C) $(\sim p \vee q)$ ☐ D) $(\sim p \wedge q)$
3. What is the value of the $\lim_{x \rightarrow 1} \frac{\sin x}{x-1}$?
☐ A) $\frac{\pi}{2}$ ☐ B) π ☐ C) $\frac{2}{\pi}$ ☐ D) Does not exist
4. If the function $f(x) = \begin{cases} \frac{\sin^2 ax}{x^2} & \text{for } x \neq 0 \\ 16 & \text{for } x = 0 \end{cases}$ is continuous at $x = 0$, then what is the value of a ?
☐ A) 16 ☒ C) 4 ☐ D) 2 ☐ B) 8
5. What is the value of $\lim_{t \rightarrow 0} \frac{\sqrt{3t^2+1}-1}{3t-7}$?
☐ A) $\frac{7}{\sqrt{3}}$ ☒ C) $\frac{1}{\sqrt{3}}$ ☐ D) $\frac{-1}{\sqrt{3}}$ ☐ B) $\sqrt{3}$
6. What is the value of $\lim_{y \rightarrow \pi} \frac{\sin y}{2 + \cos y}$?
☒ A) 0 ☐ B) 1 ☐ C) π ☐ D) 2
7. If $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ and $\mathbb{R}$ have their usual meaning, then which one of the following statements is not true?
☐ A) $\mathbb{Z} \cap \mathbb{Q} = \mathbb{Z}$ ☐ B) $\mathbb{Z} \subset \mathbb{Q}$ ☐ C) $\mathbb{R} \subset \mathbb{Z}$ ☐ D) $\mathbb{Q} \cup \mathbb{R} = \mathbb{R}$

8. What is the value of $\lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{|x|} \right)$?
☒ A) -1 ☐ B) 0 ☐ C) ∞ ☐ D) 1
9. What is the value of $\frac{dy}{dx}$, if $y = 2 - 3x^7$?
☐ A) 2 ☒ B) -3 ☐ C) 5 ☐ D) 3
10. If $A = \{1, 2, 3\}$, what is the value of $n[P(A)]$?
☐ A) 4 ☐ B) 15 ☐ C) 16 ☒ D) 8
11. What does $x \in (B - A)$ mean?
☐ A) $x \in A$ and $x \in B$ ☐ B) $x \in A$ or $x \in B$
☐ C) $x \in B$ and $x \notin A$ ☐ D) $x \notin A$ or $x \in B$

$2^3 = 8$



KATHMANDU MODEL SECONDARY SCHOOL

Set VI Science

Time: 15 Min

Subject: Com. Mathematics (0071)

SET-A

Circle the best option among the four options on your answer sheet.

GROUP 'A' MCQ (1×1=1)

Write the correct option of each question in your answer sheet.

1. If $x, y > 0$ and $\frac{1}{x} > \frac{1}{y}$, which one of the following is true?
 - A) $x > y$
 - B) $-\frac{1}{x} > -\frac{1}{y}$
 - C) $x < -y$
 - D) $x < y$
2. If $A = \{a, b, c, d\}$, what is the value of $n[P(A)]$?
 - A) 4
 - B) 15
 - C) 16
 - D) 8
3. Which one of the following is equivalent to $\sim(p \vee \sim q)$?
 - A) $\sim(p \wedge q)$
 - B) $(p \vee q)$
 - C) $(\sim p \vee q)$
 - D) $(\sim p \wedge q)$
4. What is the value of the $\lim_{x \rightarrow 1} \frac{\sin x}{1-x}$?
 - A) 1
 - B) π
 - C) $\frac{1}{\pi}$
 - D) Does not exist
5. What does $x \in (A - B)$ mean?
 - A) $x \in A$ and $x \notin B$
 - B) $x \in A$ or $x \in B$
 - C) $x \in B$ and $x \notin A$
 - D) $x \notin A$ or $x \in B$
6. What is the value of $\lim_{x \rightarrow 0^+} (\frac{1}{x} - \frac{1}{x^2})$?
 - A) -1
 - B) 0
 - C) ∞
 - D) 1
7. If N, Z, Q and R have their usual meaning, then which one of the following statements is not true?
 - A) $N \cap Q = N$
 - B) $Z \subset Q$
 - C) $Q \cup R = R$
 - D) $R \subset Q$
8. What is the value of $\frac{\Delta y}{\Delta x}$, if $y = 2x + 3$?
 - A) 2
 - B) 3
 - C) 5
 - D) 0

9. If the function $f(x) = \begin{cases} \frac{\sin^2 px}{x^2} & \text{for } x \neq 0 \\ 16 & \text{for } x = 0 \end{cases}$ is continuous at $x = 0$, then what is the value of $\sqrt{p}$?
 - A) 16
 - B) 8
 - C) 4
 - D) 2

10. What is the value of $\lim_{t \rightarrow \infty} \frac{\sqrt{2t^2+1}}{3t-5}$?
 - A) $\frac{2}{3}$
 - B) $\frac{\sqrt{2}}{3}$
 - C) $\frac{2}{\sqrt{2}}$
 - D) $\frac{\sqrt{2}}{-3}$

11. What is the value of $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x}{2 + \cos x}$?
 - A) 1
 - B) 0
 - C) π
 - D) 2

Page XI Science

Subject: Com. Mathematics (20071)

P.M.: 75
P.M.: 30

Answers are required to give their answers in their own words as far as practicable.
No figures in the margin indicate full marks.

Attempt all the questions.

GROUP 'B' [8x5=40]

12. a) Define symmetric difference of two sets. [1]
b) For what value of x , $\sqrt{x}$ is real a number? [1]
c) What are conjugates of conditional $p \Rightarrow q$? [1]
d) Define a closed interval. [1]
e) Define the contradiction of a statement. [1]
13. a) If $P \subseteq Q$, then show that $\bar{Q} \subseteq \bar{P}$. [2]
b) For any three sets X, Y and Z , show that $(X \cup Y) - Z = (X - Z) \cup (Y - Z)$ [3]
14. a) Write the antecedent and the consequent of the statement "If 5 is a natural number then $\frac{1}{5}$ is a rational number." [2]
b) Let p, q and r be any statements. Show that the following statement is true. $(p \Rightarrow q) \wedge (q \Rightarrow r) \Rightarrow (p \Rightarrow r)$ [3]
15. a) Write any two properties of inequalities. [2]
b) Solve the inequality: $-3 < 7 - 2x \leq 5$. [2+1]
16. a) For any two sets A and B , prove that $(A \cap B) = \bar{A} \cup \bar{B}$ [2]
b) If $A = [-1, 3]$ and $B = (0, 2)$, then find $A \cup B, A \cap B$ and $A - B$ [3]
17. a) Let $h(x)$ be a continuous function. What is the relationship between $\lim_{x \rightarrow p} h(x)$ and $h(p)$? [1]
b) Define derivative of $f(x)$ with respect to x . [1]
c) If $\lim_{x \rightarrow a} f(x)$ exists, what is the relationship between $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$? [1]
d) Define jump discontinuity of a function $f(x)$ at $x = a$. [1]
e) What is the value of $\lim_{x \rightarrow 0} \frac{e^{\sqrt{x}} - 1}{x}$? [1]

18. a) Evaluate the limit, if it exists. $\lim_{x \rightarrow 16} \frac{\sqrt{x} - 4}{16x - x^2}$ [2]
b) If $f(x) = \begin{cases} \frac{\sqrt{1+x} - \sqrt{1-x}}{2x+1}, & -1 \leq x < 0 \\ \frac{x}{x-2}, & 0 \leq x \leq 1 \end{cases}$ is continuous at $x = 0$. Find the value of p . [3]

19. a) Find the limit of the following function if exists: [2]
 $f(x) = \begin{cases} 2-x & \text{for } x < 2 \\ 2+x & \text{for } x \geq 2 \end{cases}$ at $x = 2$.

- b) i) Evaluate: $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x}$ [2]
ii) Is it possible to write $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = \lim_{x \rightarrow 0} x^2 \times \lim_{x \rightarrow 0} \sin \frac{1}{x}$? If not, give reason. [1]

GROUP 'C' [3x8=24]

20. a) For any two statements p and q , prove that $\sim(p \vee q) \equiv \sim p \wedge \sim q$. [2]
b) Define a Statement. How does it differ from sentence in the context of logic? Explain. [2]
c) What is the relationship between the existence of limits and the continuity of a function at a particular point? Explain. [4]

21. a) Let $f(x) = \begin{cases} \frac{\tan x - \cot x}{x - \frac{\pi}{4}}, & x \neq \frac{\pi}{4} \\ a, & x = \frac{\pi}{4} \end{cases}$ be continuous at $x = \frac{\pi}{4}$. Find the value of a . [4]

- b) Find the derivative of the function $f(x) = \sqrt[3]{(2-3x)^2}$ by using definition of the derivative. [4]

22. a) Evaluate: $\lim_{x \rightarrow y} \frac{\tan x - \tan y}{x - y}$ [2]
b) Find from the first principles, the differential coefficient of the function $\cot(\theta - a)$ [2]
c) If A and B be any two non-empty sets, verify that $A - B$ and $B - A$ are disjoint sets by choosing suitable sets A and B . [3]

Best of Luck

Class: XI 'Science'

Time: 3 hrs.

Subject: Mathematics [0071]

F.M.: 75

P.M.: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP B [8×5=40]

12. a) Define power set with an example.
 b) Define tautology with example.
 c) Under what condition does the inverse of the function exist?
 d) Write $\{x: b > x \geq a\}$ into interval form.
 e) If x is a number, what does $|x|$ geometrically represent? [1+1+1+1+1]
13. a) Show that if $A \subset B$, then $C - B \subset C - A$. [2]
 b) Let p, q, r be any statements. Show that the following statement is true,
 $(p \vee q) \Rightarrow r \equiv (p \Rightarrow r) \wedge (q \Rightarrow r)$ [3]
14. a) For any $x \in \mathbb{R}$ and $p > 0$, prove that
 $-p \leq x \leq p \Rightarrow |x| \leq p$. [2]
 b) Test whether the function
 $f: \mathbb{R} - \{3\} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x-1}{x-3}$ is bijective or not. [3]
15. a) Let $g(x) = \frac{x^2+x-6}{|x-2|}$. Find
 i) $\lim_{x \rightarrow 2^+} g(x)$ ii) $\lim_{x \rightarrow 2^-} g(x)$ [2]
 b) Evaluate: $\lim_{t \rightarrow 0} \left(\frac{1}{t\sqrt{1+t}} - \frac{1}{t} \right)$ [3]
16. a) Find from definition the derivative of $\sqrt{\tan(3-5x)}$. [2]
 b) If $(x+t)^5 = x^5 + t^5$, show that $\frac{dx}{dt} = \frac{x}{t}$. [3]
17. a) If $\lim_{x \rightarrow p} f(x)$ and $\lim_{x \rightarrow p} f(x)$ are equal, what can be concluded about
 $\lim_{x \rightarrow p} f(x)$? [3]

- b) Define jump discontinuity of the function $g(x)$ at $x = p$.
- c) What does it mean for the function f to be differentiable at $x = a$?
- d) What does $f'(x) > 0, x \in (a, b)$ geometrically represent?
- e) If $f'(x) = g(x)$, then express $f(x)$ in terms of an antiderivative of $g(x)$.

[1+1+1+1+1]

18. a) Does the function $f(x) = x^5$ have point of inflection? Explain. [2]
 b) Find the local maximum and local minimum values of the function
 $f(x) = x + \frac{25}{x}$. [3]
19. a) If $\int f(x) dx = \sin 2x + \cos^2 x + C$, find $f(x)$. [2]
 b) Evaluate: $\int \tan^5 \theta \sec^2 \theta d\theta$. [3]

GROUP C [3×8=24]

20. a) State the condition for f to be an identity function. [1]
 b) Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 5, 7, 9\}$. Given $f: A \rightarrow B$ is defined by
 $f(x) = 2x + 1$, verify that $f \circ f^{-1} = I_B$ and $f^{-1} \circ f = I_A$ [2]
 where I_X denotes the identity function on set X .
 c) Find the domain and range of $f(x) = \sqrt{x - x^2}$. [4]
21. a) Under what condition is $x = c$ said to be a critical point of the function
 $f(x)$? Also, find the critical points of $f(x) = x^{\frac{1}{3}}(4-x)$. [3]
 b) Solve: $\frac{x^3-3x}{x+2} \leq 0$. Also, represent the solution in number line. [4+1]
22. a) Is it true that a function has infinitely many antiderivatives which differ from one another only by a constant? Illustrate this statement with a suitable example. [2]
 b) What does the Squeeze Theorem say? Evaluate:
 $\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x}$ [1+2]
 c) Evaluate: $\int \frac{\sqrt{9-t^2}}{t^2} dt$. [3]

Best of Luck

KATHMANDU MODEL SECONDARY SCHOOL

SECOND TERM EXAMINATION 2082

Class: XI 'Science'

SET 'A'

Time: 25 Min

Subject: Mathematics [0071]

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

GROUP 'A' MCQ [11×1=11]

Re-write the correct option of each question in your answer sheet.

1. Which of the following compound statements is not true?

- A) $\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$ B) $\neg(p \vee q) \equiv (\neg p) \wedge (\neg q)$
C) $\neg(p \Rightarrow q) \equiv p \vee \neg q$ D) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

2. What is the solution of $|1 - x| < 5$?

- A) $(-\infty, \infty)$ B) $(-4, 6)$
C) $(-\infty, -4) \cup (6, \infty)$ D) $[-4, 6]$

3. For any two nonempty sets A and B, which of the following is true?

- A) $A \cup B \subseteq B$ B) $A \cap B \subseteq A \cup B$
C) $A \subseteq A \cap B$ D) $A \cup \bar{A} = \Phi$

4. Which of the following is the domain of the function $f(x) = \frac{1}{2x-1}$?

- A) $(-\infty, \infty)$ B) $(-\infty, \infty) - \{-\frac{1}{2}\}$
C) $(-\infty, \frac{1}{2}) \cup (\frac{1}{2}, \infty)$ D) $\mathbb{R} - \{1\}$

5. What is the derivative of $e^{\sqrt{\sin x}}$ with respect to x?

- A) $e^{\sqrt{\sin x}}$ B) $\frac{\sin x}{2\sqrt{\cos x}} e^{\sqrt{\sin x}}$
C) $\frac{\cos x}{\sqrt{\sin x}} e^{\sqrt{\sin x}}$ D) $\frac{\cos x}{2\sqrt{\sin x}} e^{\sqrt{\sin x}}$

6. In which interval is the function $f(x) = 1 - \frac{1}{x}$ is increasing?

- A) $(-\infty, \infty)$ B) $(-\infty, \infty) - \{0\}$ C) $(-\infty, 0)$ D) $(0, \infty)$

7. What is the value of $\lim_{x \rightarrow 5} \frac{x^2 - 25}{\sqrt{x} - \sqrt{5}}$?

- A) $10\sqrt{5}$ B) $20\sqrt{5}$ C) $5\sqrt{2}$ D) $-20\sqrt{5}$

8. What is the derivative of $\sin(100 - \pi)$ with respect to x?

- A) $-\cos(\pi - 100)$ B) $\cos(\pi - 100)$
C) $\cos \pi$ D) $\sin \pi$

9. What is the value of the integral $\int \frac{\cos^2(\sqrt{2}x)}{\sin(2\sqrt{2}x)} dx$?

- A) $\frac{1}{2\sqrt{2}} \ln \sec \sqrt{2}x + C$ B) $-\frac{1}{\sqrt{2}} \ln \cos \sqrt{2}x + C$
C) $\frac{1}{2\sqrt{2}} \ln \sin \sqrt{2}x + C$ D) $\frac{1}{2} \ln \sec \sqrt{2}x + C$

10. What is the value of p, if the function $f(x) = \begin{cases} \frac{x}{\ln(1+\sqrt{3}x)} & \text{for } x \neq 0 \\ p & \text{for } x = 0 \end{cases}$ is continuous at $x = 0$?

- A) $\frac{1}{\sqrt{3}}$ B) $\sqrt{3}$ C) 3 D) $\frac{1}{3}$

11. What is value of the integral $\int \sec^2(a - px) dx$?

- A) $\frac{1}{p} \tan(a - px) + C$ B) $-p \tan(a - px) + C$
C) $-\frac{1}{p} \tan(a + px) + C$ D) $-\frac{1}{p} \tan(a - px) + C$

Class: XI 'Science'

SET 'B'

Time: 25 Min.

Subject: Mathematics [0071]

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

GROUP 'A' MCQ [11×1=11]

Re-write the correct option of each question in your answer sheet.

1. What is the value of $\lim_{x \rightarrow 5} \frac{x^2 - 25}{\sqrt{x} - \sqrt{5}}$?
 A) $-20\sqrt{5}$ B) $10\sqrt{5}$ C) $20\sqrt{5}$ D) $5\sqrt{2}$
2. What is the derivative of $\sin(100 - x\pi)$ with respect to x ?
 A) $-\pi \cos(100 - x\pi)$ B) $\pi \cos(100 - x\pi)$
 C) $-\cos(\pi - 100)$ D) $-\pi \cos(\pi - 100)$
3. What is the value of the integral $\int \frac{\cos^2(\frac{\sqrt{2}x}{2})}{\sin(2\sqrt{2}x)} dx$?
 A) $-\frac{1}{\sqrt{2}} \ln \cos \sqrt{2}x + C$ B) $\frac{1}{2\sqrt{2}} \ln \sin \sqrt{2}x + C$
 C) $\frac{1}{2} \ln \sec \sqrt{2}x + C$ D) $\frac{1}{2\sqrt{2}} \ln \sec \sqrt{2}x + C$
4. What is the value of p , if the function $f(x) = \begin{cases} \frac{x}{\ln(1+\sqrt{3}x)} & \text{for } x \neq 0 \\ p & \text{for } x = 0 \end{cases}$ is continuous at $x = 0$?
 A) $\frac{1}{3}$ B) $\frac{1}{\sqrt{3}}$ C) $\sqrt{3}$ D) 3
5. What is value of the integral $\int \sec^2(-a + pt) dt$?
 A) $-\frac{1}{p} \tan(a + pt) + C$ B) $\frac{1}{p} \tan(-a + pt) + C$
 C) $\frac{1}{-a} \tan(-a + pt) + C$ D) $-p \tan(a - pt) + C$
6. Which of the following compound statements is not true?
 A) $\sim(p \vee q) \equiv (\sim p) \wedge (\sim q)$ B) $\sim(p \Rightarrow q) \equiv p \vee \sim q$
 C) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ D) $\sim(p \wedge q) \equiv (\sim p) \vee (\sim q)$

7. What is the solution of $|1 - x| < 5$?

- A) $[-4, 6]$ B) $(-\infty, \infty)$
 C) $(-4, 6)$ D) $(-\infty, -4) \cup (6, \infty)$

8. For any two nonempty sets A and B , which of the following is true?

- A) $A \subseteq A \cap B$ B) $A \cup \bar{A} = \Phi$
 C) $A \cup B \subseteq B$ D) $A \cap B \subseteq A \cup B$

9. Which of the following is the domain of the function $f(x) = \frac{1}{2x-1}$?

- A) $(-\infty, \infty) - \{-\frac{1}{2}\}$ B) $(-\infty, \frac{1}{2}) \cup (\frac{1}{2}, \infty)$
 C) $\mathbb{R} - \{1\}$ D) $(-\infty, \infty)$

10. What is the derivative of $e^{\sqrt{\sin x}}$ with respect to x ?

- A) $\frac{\cos x}{2\sqrt{\sin x}} e^{\sqrt{\sin x}}$ B) $e^{\sqrt{\sin x}}$
 C) $\frac{\sin x}{2\sqrt{\cos x}} e^{\sqrt{\sin x}}$ D) $\frac{\cos x}{\sqrt{\sin x}} e^{\sqrt{\sin x}}$

11. In which interval is the function $f(x) = 1 - \frac{1}{x}$ is increasing?

- A) $(-\infty, 0)$ B) $(0, \infty)$
 C) $(-\infty, \infty)$ D) $(-\infty, \infty) - \{0\}$

KATHMANDU MODEL SECONDARY SCHOOL

FIRST TERM EXAMINATION 2082

Group: XI Science

SET 'A'

Subject: Computer Science (4273)

5 M., 50

Students are required to give their answer in their own words as far as practicable.

Write in the margin indicated full marks:

MULTIPLE CHOICE QUESTIONS.

GROUP 'A' (9×1=9)

1. _____ are numbers and encoded characters generally used as operands.
a. Input b. Data c. Information d. Stored values
2. Loading Computer works on the supply of:
a. Physical strength b. Discrete signals
c. Magnetic strength d. Continuous electric pulse
3. Which is the major component of first generation computers?
a. Transistor b. Microprocessor
c. Vacuum Tube d. IC
4. A bar code reader is an example of _____ device.
a. processing b. storage c. input d. output
5. What is the binary equivalent of a decimal number 23?
a. 10101 b. 10100 c. 11010 d. 101101
6. Which of the following isn't positional number system?
a. Binary b. Octal c. Decimal d. Roman
7. A light sensitive device that converts drawings, printed text or other images into digital form is
a. Keyboard b. Scanner c. Plotter d. OMR
8. A magnetic tape is a type of _____ access storage device.
a. Sequential b. Direct c. Strip d. Indirect
9. What is the maximum Zoom size in MS word?
a. 150 b. 200 c. 500 d. 350

GROUP 'B' (5×5=25)

SHORT ANSWER TYPE QUESTIONS.

10. Why we use multimedia? Define linear and non-linear multimedia with examples. Explain the different components of multimedia. [3+3+5]
11. Explain the memory hierarchy in a computer system with a neat and labeled diagram. Describe the different benefits of memory and discuss their characteristics in terms of speed, cost, and capacity. [2+2+3]
12. Describe any three types of positional number system. Convert the following as indicated. [3+3+3]

a. $(93.5)_{10} = (?)_2$

b. $(8AD.DA)_{16} = (?)_{10}$

Subtract 1011 from 10101, using 1's and 2's complement methods.

13. What is mobile computing? Suppose the internet and mobile networks suddenly stopped working for one full week. Describe at least two problems people would face in education, business, and daily life. Suggest at least two alternative solutions people could use during that time. [3+3+3+3]

OR

Explain in detail the concept of mail merge in Microsoft Word. What are its primary benefits and disadvantages? [5]

14. Define output devices with examples. Write their functions. Write any four differences between CRT monitor and LCD monitor. [3+2+2]

GROUP 'C' (2×8=16)

LONG ANSWER TYPE QUESTIONS.

15. What do you mean by computer architecture? Describe each functional components of computer architecture with typical diagram. [3+7]
16. You are given to prepare a project report in MS Word about "Generations of Computers."

- a. Explain how you would use any four Word features (like Table of Contents, Page Numbering, WordArt, Styles, SmartArt, etc.) to organize the report neatly. [4+1+5=6]
- b. Write why these features are useful for readers. [2]

"Best of Luck"



KATHMANDU MODEL SECONDARY SCHOOL

FIRST TERM EXAMINATION 2082

Class: XI Science

SET 'B'

Time: 2 hrs.

Subject: Computer Science (4221)

F.M.: 50
P.M.: 20

Candidates are required to give their answer in their own words as far as practicable.
No figure in the margin indicated full marks.

Attempt all the questions.

GROUP 'A' (9x1=9)

MULTIPLE CHOICE QUESTIONS.

- The three main components of computer are
a) CPU, Memory, Tape
b) I/O, Printer, Mouse
c) CPU, I/O, Memory
d) Tape, I/O, Floppy disk
- Which type of computer is smaller than a mainframe but larger than a microcomputer, and used by medium-sized organizations?
a) Hybrid Computers
b) Super Computers
c) Mini Computers
d) Analog Computers
- Which is the major component of fourth generation computers?
a) Transistor
b) Microprocessor
c) Vacuum Tube
d) IC
- A speaker is an example of device.
a) processing
b) storage
c) input
d) output
- What is the binary equivalent of a decimal number 31?
a) 10101
b) 11111
c) 11010
d) 101101
- How many bits are there in 2.5 kilobytes?
a) 20,480 bits
b) 25,600 bits
c) 40,960 bits
d) 20,480 bytes
- The maximum number of bits used to represent hexadecimal number in binary is
a) 3
b) 4
c) 8
d) 16
- The is a small input device used to select and display objects on a screen.
a) Ink
b) Magnet
c) Light pen
d) None of these
- Which of the following is not the font style?
a) Bold
b) Italics
c) Regular
d) Superscript

GROUP 'B' (5x5=25)

SHORT ANSWER TYPE QUESTIONS.

- Why we use multimedia? Define linear and non-linear multimedia with examples. Explain the different components of multimedia. [1+1+3]
- Explain the memory hierarchy in a computer system with a neat and labeled diagram. Describe the different levels of memory and discuss their characteristics in terms of speed, cost, and capacity. [2.5+2.5]
- Describe any three types of positional number system. Convert the following as indicated. [3+1+1]
a) $(93.6)_{10} = (?)_2$
b) $(BAD.DAD)_{16} = (?)_{10}$

OR

- Subtract 1011_2 from 10101_2 using 1's and 2's complement methods. [2.5+2.5]
- What is mobile computing? Suppose the internet and mobile networks suddenly stopped working for one full week. Describe at least two problems people would face in education, business, and daily life. Suggest at least two alternative solutions people could use during that time. [1+1+1+1+1]

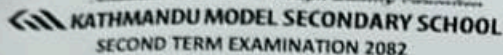
OR

- Explain in detail the concept of mail merge in Microsoft Word. What are its primary benefits and disadvantages? [5]
- Define output devices with examples. Write their functions. Write any four differences between CRT monitor and LCD monitor. [1+2+2]

GROUP 'C' (2x8=16)

LONG ANSWER TYPE QUESTIONS.

- What do you mean by computer architecture? Describe each functional components of computer architecture with typical diagram. [1+7]
- You are given to prepare a project report in MS Word about "Generations of Computers."
a) Explain how you would use any four Word features (like Table of Contents, Page Numbering, Wordart, Styles, SmartArt, etc.) to organize the report neatly. [4x1.5=6]
b) Write why these features are useful for readers. [2]



SECOND TERM EXAMINATION 2082

чрез: XI "Science"

SET "B"

type: 2 hrs.

Subject: Computer Science (4271)

F.M.: 50

P.M.: 20

Candidates are required to give their answer in their own words as far as practicable. The figure in the margin indicated full marks:

attempt all the questions.

GROUP 'A' [9×1=9]

Multiple Choice Questions.

1. The term GIGO is related to which characteristics of computers?
 - a) Speed
 - b) Automatic
 - c) Accuracy
 - d) Reliability
2. NAND gate means
 - a) Inversion followed AND gate
 - b) AND gate followed by an inverter
 - c) AND gate followed by an OR gate
 - d) None of above
3. Which software is used to help analyze, configure, optimize and maintain the computer?
 - a) Device driver
 - b) Utility software
 - c) Application software
 - d) All
4. Which symbol must all formula begin within MS-Excel?
 - a) =
 - b) +
 - c) %
 - d) @
5. The numeric data entered in the worksheet is _____ aligned by default.
 - a) Left
 - b) Right
 - c) Center
 - d) Top
6. Which HTML tag is used to create smallest heading in a webpage?
 - a) <h0>
 - b) <h3>
 - c) <h6>
 - d) <h7>
7. JPEG stands for _____.
 - a) Joint photographic Exports Group
 - b) Joint Physical Experts Group
 - c) Joint Physical Exports Group
 - d) Joint Photographic Experts Group
8. Microsoft Windows is an example of?
 - a) Open source OS
 - b) Proprietary OS
 - c) Both
 - d) None
9. Programs stored in the ROM are called as:
 - a) Firmware
 - b) Formware
 - c) Farmware
 - d) Firmware

GROUP 'B' (5×5=25)

Short Answer Questions

10. Justify the statement: "The mechanical nature of forming characters means that **impact printers** generate much higher noise levels during operation, which makes them unsuitable for quiet environments where **non-impact printers** are the preferred choice."

11. Explain the primary memory with its type. [5]
12. Differentiate between **System Software** and **Application**.

'OR'

Differentiate between Linear Multimedia and Non-Linear Multimedia. [5]

13. Write HTML code to print following:
- a. First Generation
 - i. Vacuum Tube
 - ii. Machine Language
 - iii. ENIAC
 - b. Second Generation
 - i. Transistors
 - ii. Assembly Language
 - iii. IBM 1401
 - c. Third Generation
 - i. Integrated Circuit
 - ii. High Level Language
 - iii. PDP-8
14. In Excel, why we use Cell referencing? Explain Absolute Cell Referencing and Relative Cell Referencing.

GROUP 'C' [2×8=16]

Long Answer Questions

15. What do you mean by Operating system? Explain any 6 types of operating system on the basis of **Processing Mode**. [2+6]
16. Explain all Logic Gates with statements, Boolean expression, logical symbol, truth table and Venn-diagram.

Best of Luck



Knowledge, Skill, Power, Through Quality Education

KATHMANDU MODEL SECONDARY SCHOOL

SECOND TERM EXAMINATION 2082

Class: XI 'Science'

SET 'A'

F.M.: 50

Time: 2 hrs.

Subject: Computer Science (4271)

P.M.: 20

Candidates are required to give their answer in their own words as far as practicable.

the figure in the margin indicated full marks:

Attempt all the questions.

GROUP 'A' [9×1=9]

Multiple Choice Questions.

- Storage which stores or retains data after power off is called _____.
a) Volatile storage b) Non-volatile storage
c) Cache memory d) None of above
- Which logic gate is represented by the Boolean function $F = A \cdot B + A' \cdot B$?
a) XOR b) XNOR c) AND d) OR
- Which software is used to perform specific task?
a) System software b) Application software
c) Utility software d) None
- Which one of the following font-size can not be selected from tool bar?
a) 11 b) 12 c) 13 d) 14
- What is the keyboard shortcut to start a PowerPoint presentation from the beginning?
a) F5 b) Ctrl + F5 c) Alt + F5 d) Shift + F5
- Which of the following tags do you use to create an ordered list?
a) b) c) d) <LIST>
- A video consists of a sequence of _____.
a) Signals b) frames c) packets d) slots
- Linux, Unix etc. are the examples of?
a) Proprietary OS b) Open source OS
c) Both d) None

- Which type of memory can speed up computer processing?
a) ROM b) Cache memory
c) Registers d) Both A and B

GROUP 'B' [5×5=25]

Short Answer Questions.

- Justify the statement: "The mechanical nature of forming characters means that impact printers generate much higher noise levels during operation, which makes them unsuitable for quiet environments where non-impact printers are the preferred choice." [5]
- Explain the primary memory with its type. [5]
- Differentiate between System Software and Application Software. [5]

'OR'

Differentiate between Linear Multimedia and Non-Linear Multimedia. [5]

- Write HTML code to print following:

- First Generation
i. Vacuum Tube
ii. Machine Language
iii. ENIAC
- Second Generation
i. Transistors
ii. Assembly Language
iii. IBM 1401
- Third Generation
i. Integrated Circuit
ii. High Level Language
iii. PDP-8

- In Excel, why we use Cell referencing? Explain Absolute Cell Referencing and Relative Cell Referencing. [1+4]

GROUP 'C' [2×8=16]

Long Answer Questions.

- What do you mean by Operating system? Explain any 6 types of operating system on the basis of Processing Mode. [2+6]
- Explain all Logic Gates with statements, Boolean expression, logical symbol, truth table and Venn-diagram.

"Best of Luck"

- d) Many teenagers use drugs as a way to escape pain, confusion, and loneliness.
 I. happiness II. misery III. success
 e) Teenagers may first be exposed to drugs at and discos.
 I. schools II. parties III. homes

Write short answers to the following questions in about 75 words each.

- a) Explain the significance of the seasons in the story, *The Selfish Giant*. [5x2=10]
 b) How is the poem, 'A Red, Red Rose' a romantic poem?

OR

Why is it necessary for the younger generation to take responsibility for carrying on traditions. Discuss. (*Sharing Traditions*)

- c) The poem, *Corona Soys* reflects ecological awareness. Discuss.

- d) What does the line "*The Earth is not your property alone*" mean? (*Corona Soys*)

OR

What is a hyperbole? Explain any two hyperboles used in the poem, "A Red, Red Rose."

- e) What did the Giant build around his garden? Why? (*The Selfish Giant*)

Write long answers to the following questions in about 150 words each.

[2x5=10]

- a) The poem, *A Red, Red Rose* celebrates love as eternal and unchanging. Discuss.

OR

Why is it difficult to transmit the oral traditions in the modern world of globalization and technology? Describe. (*Sharing Tradition*)

- b) What moral lessons do we learn from "The Selfish Giant"? Explain.

Write a few paragraphs either in favour of or against the ban on social media. [7]

Write a letter to your cousin who is currently working in Korea encouraging him to return to Nepal and use the valuable skills and experiences he has gained abroad for the development of the nation. Also, remind him of the growing opportunities at home and the importance of contributing to the progress of the homeland instead of staying away for a long time. [8]

Traditionally, politics has been dominated by older leaders, believed to have the wisdom and experience to govern. Recently, many young people want to participate in politics, claiming they bring energy, innovation, and fresh ideas. This raises a debate: should young people take an active role in

politics, or should leadership remain with the experienced elderly. Write an essay discussing the advantages and disadvantages of allowing young people to involve in politics. Support your answer with examples and give your opinion. [10]

7. Do as indicated:

[10x1=10]

- a) Talk to the children (polite/politely/polliteness). (Choose the best option)

- b) I was **shocked** to hear the news. (Identify the word class of the bold letter word)

- c) Mina will pass the exam surely. (Rewrite the sentence putting the adverb in correct place)

- d) He often visits us. / Often he visits us. (Choose the best option)

- e) There is a book (onto/into/on) the desk. (Choose the best option)

- f) The cat jumped the bar and landed in the field. (Fill in the blank with a suitable preposition)

- g) She (seek) advice from her teacher before making the decision. (Rewrite the sentence in simple past tense)

- h) I think she (pay) the bill tomorrow. (Rewrite the sentence using correct tense)

- i) The monkey on the circus rope has lost the balance. It (fall) on the ground. (Rewrite the sentence using correct tense)

8. Choose the best options: [5x1=5]
 j) If I (be) you, I'd visit the doctor. (Rewrite the sentence using correct tense)

a) The new rules were mandatory for all students. The meaning of the word **mandatory** is:

- I. optional II. compulsory III. flexible IV. unnecessary

- b) Which word comes first, if **gate**, **gain**, **game**, **garden**, **gas** are arranged in alphabetical order?

- I. gate II. gas III. gas IV. garden

- c) The abbreviation **v.** in a dictionary stands for:

- I. verb II. value III. verify IV. various

- d) She looked at me when I told the truth.

- I. angrily II. angry III. anger IV. angrier

- e) The prefix **sub-** in the word **submarine** means:

- I. above II. under III. opposite IV. again

Culture and tradition are the true heartbeat of our identity. Let's celebrate them in an authentic way and commit to preserving them.

Happy Thari!

* Best of Luck *

gene: 3 hrs.

1999

F.M.: 75

P.M.: 30

Read the following text and do the activities that follow

as also |

persuade them from using it.

[5x1=5]

- from using drugs

 $[5 \times 1 = 5]$

[5x] = 1

- ### III. hobby

- b) We the money electronically to confirm the reservation.
- c) The child was healthy
- d) The classroom was with new interactive whiteboards.
- e) Her grandmother's wedding ring was the most item she owned.

2. Based on the literary texts in your book, write short answers to Any Five of the following questions in about 75 words each. [5×2=10]

- a) A rose is a fragile, short-lived flower, yet the poet claims his love will last until the "seas gang dry." How does Burns bring together the temporary nature of a rose with the eternal nature of his love? (*A Red, Red Rose*)
- b) LaPena emphasizes that tradition is not just about facts, but about the spirit of the people. How does sharing a story transfer this spirit from one generation to the next? (*Sharing Traditions*)
- c) Why does the Giant build a wall around his garden? What does this wall represent in terms of his emotional state and his relationship with society? (*The Selfish Giant*)
- d) Jobs argues that you cannot connect the dots looking forward, only looking backward. How does his decision to 'drop out' of college and 'drop in' on a calligraphy class illustrate it? (*How to Live before you Die*)
- e) E.E. Cummings is famous for his unique use of punctuation. Why has he used lowercase in the poem? (*who are you little i*)
- f) The virus claims that it did not come of its own accord but was 'invited.' What human actions is the poet referring to as the causes of 'invitation'? (*Corona Says*)
- g) The title *Trifles* suggests small, insignificant things. How do these "trifles" help solve the murder mystery? Discuss. (*Trifles*)

3. Based on the literary texts in your book, write long answers to Any Two of the following questions in about 150 words each. [2×5=10]

- a) When does passion become poison? How does the painter's devotion to his art blind him to his moral obligations as a husband? Discuss how his passion turns into a destructive obsession in the story, *The Oval Portrait*?
- b) The women hide the evidence from the men. Why do they do this? Discuss. (*Trifles*)
- c) In the story, *The Wish*, imagination is not an escape but a trap. Roald Dahl's turns the boy's imagination of a simple carpet into a life-or-death situation. How does Dahl blur the line between the boy's imagination and reality? (*The Wish*)

4. Nepali society, traditionally known for its close-knit family structures, is now experiencing a weakening of these bonds. What do you think are the

key factors contributing to this change in Nepal, and what measures could be taken to address it? Write a couple of paragraphs on it [7]

5. Imagine you are a retired Gurkha soldier living in the UK. Write a nostalgic letter to your childhood friend who still lives in your hometown in eastern Nepal, Jhapa recalling about your school days, sharing stories of your life abroad, and expressing your dream to return home to build a community library. [8]

6. These days, weddings in Nepal are famous for their beauty, luxury, and grand celebrations. Along with this, they also involve a lot of waste, extravagance, and showy use of money. Write a diary entry about your experience of attending real/imaginary such a wedding. [10]

7. Do as indicated: [10×1=10]

- a) She cannot walk that far. In this sentence, the word 'that' is (a determiner / a pronoun / an adverb) (Choose the correct option)
- b) She will definitely help us. / She definitely will help us. (Choose the correct sentence)
- c) The bird flew (off/out of/ up) the cage. (Choose the correct option)
- d) He (sees/sawed/saw) a movie last night. (Choose the correct option)
- e) They (will/ are going) to visit Jhapa next week. They have already booked the tickets. (Choose the correct option)
- f) The police (has/have/is) arrested the culprit. (Choose the correct option)
- g) The bird flew away. The underlined verb is (a transitive/ an intransitive/a linking) verb. (Choose the correct option)
- h) We (meet/will meet/are meeting) her tomorrow at 7:30 AM. (Choose the correct option)
- i) You (should / had better) start revising the lessons now. Otherwise, you might fail the exam. (Choose the correct option)
- j) He was not so studious. He passed the exam. (Join the sentences with 'Despite')

8. Rewrite the following sentences choosing the best options: [5×1=5]

- a) She spoke with (confidence / confident / confidently).
- b) They had a (quick / fast / speedy) look at the report.
- c) He was murdered in cold blood. It means he was killed (keeping in freeze / deliberately/ cruelly)
- d) The prefix that does not go with the word 'place' is: (mis- / re- / un-)
- e) Please turn (off / out / down) the volume; it's too loud.

Best of Luck

KATHMANDU MODEL SECONDARY SCHOOL
SECOND TERM EXAMINATION 2082

Class: XI 'Science'

Time: 3 hrs.

Subject: English [0031]

F.M.: 75

P.M.: 30

Candidates are required to give their answer in their own words as far as practicable. The figure in the margin indicated full marks.

Attempt all the questions.

Read the following TEXT and do the activities that follow.

Photo Perfect Travel Cameras

Thank you for purchasing a Photo Perfect Travel Camera – a camera that will capture all your precious travel moments! We have made every effort to make our cameras robust and features user-friendly so that you get the very best while you are travelling. This Rapid Guide will introduce you to some of the basic components and features of the camera and how to use them. Please read the Extra User Guide carefully for more in-depth instruction before using the camera. Be sure to keep the Rapid Guide handy.

Battery

To guard against possible leakage, we do not install batteries. To begin operation, insert 4 AA batteries into the battery compartment. Standard batteries will last for up to 12 hours of continuous use. Lithium batteries will last for 24 hours continuously.

Lens

Your Photo Perfect camera is equipped with an 80x digital zoom wide angle lens with image stabilizer. The camera is also fitted with our patented Airwalker JJTM technology enabling you to cushion zoom. Avoid touching the lens and always clean with a cotton cloth.

Self-timer

With our self-timer technology, you can appear in your own photograph! The camera's shutter can be delayed by 5 to 10 seconds once the shutter button has been set. Before activating the self-timer make sure the camera is placed on a flat surface. The yellow border in the view finder determines what will appear in your picture.

Image quality and size

Images are stored in your camera's memory stick in a compressed format, which allows you to store more photographs. Due to compression, the preview of the images on the camera's display screen may reveal a loss of detail. When the images are printed or transferred to other high-resolution viewing devices (such as a computer), full resolution will be displayed.

Setting the right image size is important to get a perfect printout of your photographs. You can choose from manual and auto settings. The Photo Perfect user-friendly image setting button allows you to select any of the three size setting options. For an actual printout of a 4 x 6 inch photo, set the image button to 'A' on the dial. For a printout of a 7 x 9 inch photo, set the button to 'B' and for a printout of a 10 x 12 inch photo, turn the dial to 'C'. Note: A larger image size setting will require more memory.

Storage

If the camera is not going to be used for more than 2 weeks, please remove the batteries. Store the camera in its supplied case away from magnetic fields created by other electronic equipment (TVs or radios) and within a temperature range of 5°C to 35°C.

Accessories

Photo Perfect cameras come with a standard 4GB memory card. A range of other accessories are available at a reasonable cost, which can be chosen based on user requirements:

- Memory cards of 15GB and 30GB.
- 4 rechargeable AA batteries and charger.
- Genuine leather camera cover, upgrade from standard inclusion cloth cover.
- Handy photo printer weighing only 750 g.

A. Answer the following questions.

[5×1=5]

- a) How long will standard AA batteries last, if it is used nonstop?
- b) What is the special technology that enables the camera to cushion zoom called?
- c) Which image setting should be used for a 7 x 9 inch printout?
- d) Where should you store the camera to protect it from magnetic fields?
- e) What does the yellow border in the view finder mean?

B. Read the text again and write

[5×1=5]

True if the statements agree with the information given in the text.

False if the statements contradict the information given in the text.

Not Given if there is no information in the text.

- a) In order to protect the camera, batteries are not fitted.
 - b) Lithium batteries are recommended for the camera.
 - c) The 'C' setting uses more memory than the 'A' setting.
 - d) The camera should not be used in a very cold place.
 - e) The buyer should pay extra for the leather cover.
- C. Look at the UNDERLINED words in the text. Complete these sentences using them. Two of them are not necessary.**
- a) A sudden thunderstorm the flight for two hours.

[5×1=5]

वृक्षारोपण गर्नुपर्दछ । खोलाका किनारामा तटबन्ध बाँध्नुपर्दछ । वन अङ्गणको वरिपरि गर्नुपर्दछ । खोला किनारबाट आर्षाधिक बाढुवा, मिटी निकाल्नुहुँदैन । पछिरो जाने ठाउँमा पातीको निकास छोड्नुहुँदैन । पछिरो जाने क्षेत्रमा वर्षाको समयमा खेतीपाती पनि गर्नुहुँदैन । खेतीपातीले गर्दा माटो कम्यो भई पछिरो जाने भन्नु बढी सम्भावना रहन्छ । आर्षाधिक जलसङ्ग्रहमा बुझि, वातावरणीय जलमूलनको अभावमा जलशक्ति बर्बाद तथा खडेरी पर्ने सम्बन्ध । यस विषयमा हामी मानवले सजग र सचेत भएर सोच्न जरुरी छ । त्यस्तै वातावरणीय जलमूलनको अभावमा किमपात हुने र हुरीमातस आउने गर्छ । यसमा प्राकृतिक प्रकोपबाट बच्न, आउने सम्भावित क्षतरबाट बच्न सुरक्षित बसाइको व्यवस्था गर्नु जरुरी छ । यदि बसाइ सुरक्षित भएन भने मानवको जीवन र धनको क्षति हुन सक्छ ।

धनरहक :

क) मानिसले चाहिने प्राकृतिक प्रकोपलाई के गर्न सक्छ ?

(अ) बाढामा राख्न (आ) त्यनीकरण गर्ने (इ) चेबास्ता गर्ने (ई) लेखाबोखा गर्ने

ख) तटबन्ध बनेको के हो ?

(अ) खोला लेस्ने काम (आ) खोला किनारमा वृक्षारोपण गर्ने काम

(अ) खोला किनारमा पक्का लगाउने काम (ई) खोलाबाट अर्कोतिर फर्काउने काम

ग) खोला किनारबाट मानिसहरूले के निकाल्नु हुँदैन ?

(अ) मोटोबस्त्र (आ) पैनी अथवा नहर (इ) पानीको निकास (ई) बाढुवा तथा मिटी

घ) आर्षाधिक वर्षा भनेको के हो ?

(अ) अतिबृष्टि (आ) अनाबृष्टि (इ) सघनबृष्टि (ई) अन्यबृष्टि

16. तलको बन्नुछेदबाट चारबोटो बुझा टिपी एकतृतीयांशमा सारांश लेख्नुहोस् : (४)

विश्व स्वास्थ्य सङ्गठनको परिभाषाअनुसार राखी स्वास्थ्य भनेको रोग वा दुर्बलताको अभाव मात्र नभएर शारीरिक, मानसिक र सामाजिक रूपले पूर्ण तन्त्रुस्तीको अवस्था हो । जुन देशका नागरिकहरू स्वस्थ हुन्छन्, त्यस्तो देशको प्रगति इततर हुन्छ । कुनै पनि देशको स्वास्थ्य स्थिति कति सबल छ भन्ने कुरा त्यहाँका नागरिकहरूमा सुर्तीजन्य पदार्थको प्रयोगको स्थिति, उच्च रक्तचाप तथा मोटोपनको स्थिति । स्वास्थ्यमा जोखिम, शुद्ध पानीको उपलब्धता, औसत आयु, कुपोषणप्रस्त जनसङ्ख्याको प्रतिशतका साथै दुर्घटना, इन्ड वा प्राकृतिक विपत्ति जस्ता अन्य कारणबाट हुने मृत्युको अवस्था अरिमा निर्भर हुन्छ । जस्ता सुचकका आधारमा संसारकै सबैभन्दा राम्रो स्वास्थ्य स्थिति भएको देशको रूपमा स्वेनलाई लिइन्छ । सन् १९०० वर्षको इतिहास भएको चिकित्सा विज्ञान मानव सभ्यताको महत्वपूर्ण कान्छो उपलब्धि हो र यसको मुख्य उद्देश्य व्यक्तिको शारीरिक तथा मानसिक स्वास्थ्यलाई उच्चतम चिन्तुमा पुऱ्याउनु हो । स्वास्थ्य विज्ञानमा रोकथाम, निदान र निवारणको तीनबोटो विधिमा ध्यान दिइन्छ । यी विधिमध्ये चिकित्सा विज्ञान मुख्यतया निदान र निवारणमै अभिमुखी छ । रोग लागेपछि उपचार गर्नुभन्दा रोग लाग्न नदिनु नै उत्तम हो भन्ने उक्तिलाई कम महत्व दिइएको छ । अर्थात् चिकित्सा विज्ञानमा रोकथामलाई बढी रोग लाग्न नदिने उपचार गर्ने परिपाटीले बढी महत्व पाएको छ यद्यपि पछिल्लो समयमा चिकित्सा विज्ञानको खोप, जन्मतिमा जारी हुने विभिन्न जनस्वास्थ्यसम्बन्धी सुचना, जानकारी, तालिम आदिको व्यवस्थाले चिकित्सा विज्ञान रोकथामको गतिविधिमा पनि तालिम पाएको छ ।

17. कुनै एक धनको उत्तर दिनुहोस् : (४)

क) चिरामी पोरामे विद्यालय आउनु सलको कारण देखाउँदै एक दिनका लागि विद्या माग्नै तपाईंको विद्यालयका प्राध्यापलाई एउटा निवेदन लेख्नुहोस् ।

ख) आफ्नो उद्योगमा उत्पादित हर्बल चिन्ताको प्रचार प्रसारका लागि बनाइने एउटा व्यापारिक विज्ञापनको नमुना तयार गर्नुहोस् ।

18. कुनै एक धनको उत्तर दिनुहोस् : (४)

क) तपाईं अध्ययनरत विद्यालयले आफ्नो शारिक उत्सव कार्यक्रम मनाउँदा गरेका सम्पूर्ण कार्यक्रमलाई समेटेर प्रतिवेदन तयार गर्नुहोस् ।

ख) बढ्दो बढ्दै दुर्घटना शीर्षकमा १२४ शब्दसम्मको टिप्पणी लेख्नुहोस् ।

19. जातीय विषयको जलन कमरी गर्न सकिन्छ ? १२४ शब्दसम्ममा आफ्नो प्रतिक्रिया लेख्नुहोस् : (४)

19. कुनै दुई धनको उत्तर दिनुहोस् : (४+४=८)

क) विद्यार्थी विषयमा पढी सीधिएका धनहरूको उत्तर दिनुहोस् :

संस्कृति हाथो परिचय हो र हाथो अनुभूति हो । संस्कृति हाथो बोल्ने, विचार गर्ने र सोच्ने तरिका पनि हो । हाथो संस्कृति बोल्नुपर्छ, त्यस्तै हाथो संस्कृति अनुभवकायनुपर्छ । हामी हाथो संस्कृति हो, मिहार हाथो संस्कृति हो । गिज, मुडगुर्गमा, उजौरी, उजौरी, जोहार, चितवन, इट सवै हाथो संस्कृति हुन् । सभ्यताको यस भाषामा म बारम्बार सोझु र सोझु यस्ता सुन्दर जलसलाई हाथीले सधुपयोग गर्नेछ । संस्कृतिका नाममा लिइए, बर्ग, जातिका आधारमा हुने भेद चिक्छि हुन् । ती कुरा हट्नुपर्छ र सीधियानले यसको सुनिश्चिता पनि गरेको छ ।

धनरहक :

अ) संस्कृति भनेको के हो ?

आ) सीधियानले सुनिश्चित गरेका संस्कृतिक विषयहरूको के हुन् ?

ब) विद्यार्थी कथा पढी सीधिएका धनहरूको उत्तर दिनुहोस् :

खोला अघेल फेरि गाउँतिर सोधिएको रहेछ । यो खोलाते पनि गाउँलाई पुग्नेको स्वाधो । हुन त स्वाधो मान्ने बन्नु खोलाप्रति अन्याय ठहर्छ । यो खोलाते गाउँलाई पानेको पनि छ । हाथो विगतको लगभग आधा उमेर यही खोलाको दिन र बगरहरूमा बितेको थियो । कहिले ओरडा खाने निहुँमा त कहिले गाउँनै चराउने निहुँमा । खोलाका बाढी आएको बेला हाथी गाउँको पुष्कर समार बाढी लने गर्थ्यो अथवा यस्तो बेला हाथी खोलाको किनारमा बसेर बाढीको आनन्द म्छ्थ्यो । तन डराउने कोही मानिस आए भने हाथी तीनचार जना एकजनाको हात समाउँदै बाढीमा होलिनथ्यो र किनारमा पस्नेहरूको मानिसलाई तारिदिन्थ्यो र यस्तो काममा खास गरी परिछनकै विशेष स्थिति थियो । 'चर्चिण कथा छ, गए सातको वर्षामा मृत फुटोको एकबन चिसो र मिठो पाती छ ।' ऊ पाती लिएर आइसकेको थियो ।

धनरहक :

अ) खोलाते स्वाधो भन्ने कुरा मात्र गर्दा कसरी अन्याय हुन्छ ?

आ) 'म' पाइका खोलासँगका वासजनुभूति कस्ता छन् ?

ग) विद्यार्थी कथा पढी सीधिएका धनहरूको उत्तर लेख्नुहोस् :

बल सानीका आँखा उठे । त्यसलाई आफ्नो अङ्गप्रत्यङ्ग कसो बुझिन लागेको छैन थोडेर बितेको दीर्घ समयको सम्झना थियो र भन्नु प्रत्यक्षमा त्यसले देखे, बाबाको तीन चौधै पूरा भएपछि त्यसले पाइछ - आमा समाहितको साथसाथै जीवन सध्या पनि आइपुगेको रहेछ । दुनियाँ जकै भएको रहेछ, यस वर्तमानसँग त्यस सानीका कुनै प्रकारको सामन्जस्य रहेनछ । बाबाते अकस्म्य मुखलाई घोडा दिइरहेको थियो, सानी सिधिल, काला भै हुन गई । उभिरधनमा शरीर साई नै गलेको जस्तो भाव हुन लाग्यो र त्यसै झरमा बचक बत्ती ।

धनरहक :

अ) सानीको प्रतीक्षा कसरी समाप्त भयो ?

आ) यस अनुच्छेदमा सानीको कस्तो मनोविज्ञान प्रस्तुत भएको छ ?

19. कुनै एक धनको उत्तर दिनुहोस् : (८)

क) 'वीर पुर्वा' कवितामा कविले हाथो पुर्वाको योगदानलाई कसरी सम्झिएका छन् ? समीक्षात्मक उत्तर लेख्नुहोस् ।

ख) विषेदरहित समाजको स्थापनामा योगमायाते विद्यो योगदानको चर्चा गर्नुहोस् ।

19. विद्यार्थी मध्ये कुनै एक शीर्षकमा २५० शब्दसम्मको निबन्ध लेख्नुहोस् : (८)

क) मलाई मन परेको पुस्तक

ख) अस्थिर राजनीतिगत अवस्थामा पारेको प्रभाव

ग) नेपालमा पर्यटन

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अक्षा : ११ (विज्ञान सहकार्य) पूर्णांक : ७५
 समय : १ घण्टा विवरण : नेपाली (००११)
 उपरीणांक : ३०
 तल्लोको अवलोकन गरी तलका उल्लेख गरिनुपर्ने छ । तलका विषयमा तिरैको अङ्कले पूर्णांक लगाउनेछ ।

प्रश्नहरू अनिवार्य छन् ।

१. विद्युत् बलुच्चैरबाट रेखाङ्कित वर्षाहरूको उत्पत्त्य प्रत्यक्ष र प्रात्यक्ष छुट्याई लेख्नुहोस् : (३)
 २. पञ्चासी माथ लाग्ने आघाटी जग्गालाई बारी भनिन्छ । बारीमा धान नउब्जाए पनि मकै, कोदो, धान लगायतका खाद्यान्न उब्जाइन्छ ।
 ३. हृदय पुनर्निर्माण गर्नुहोस् : (४)
 यदि विट्ने गदा लगभग उदास र आँई उसको स्वर मेरो कानमा गुण्गो उसले जसरी अघ्यारो मुन्नुमा बा विनाभुनिक मलाई यो खबर सुनायो ।
 ४. तलको बलुच्चैरमा रेखाङ्कन गरिएका शब्दको परवर्ग पहिचान गरी लेख्नुहोस् : (२)
 क. एकमात्र प्रेम गरेर मत्तको निम्ति बलबल्लाई दिन्थी तर मसँग रमाउन मान्निताउन चाहन्थी ।
 ५. विद्युत् प्रणालीको उत्तर दिनुहोस् : (२+२=४)
 क। तलको बलुच्चैरबाट दुई ठुलो तत्त्व र आणवुक्त शब्द थोडी लेख्नुहोस् :
 एकाइसौ शताब्दी भनेको विद्युतीय शताब्दी हो । आज हामी कम्प्युटरको युगमा छौ । यस युगले विद्युतीय क्षेत्रको कार्य कम्प्युटरबाट गरिरहेछ ।
 - ख। विद्युत् शब्दहरूलाई शब्दकोशीय रूपमा मिलाई लेख्नुहोस् :
 घन, औषध, हरियो, ज्ञान
 ६. विद्युत् बलुच्चैरबाट तीनभेटा पारिभाषिक शब्दहरू पहिचान गरी बाक्वमा प्रयोग गर्नुहोस् : (३)
 कास्माडी उचायकाको मिठमाइ र घुलो र धुपामाई छिचोले पृथ्वी राजमार्गको बाटो हुँदै रारा पुगेर त्यहाँको दुरवासलेकन गराईको अनुभूतिने त जो कोहीलाई कबिता लेख्न सक्ने बनाउँदो रहेछ ।
 'अबसा'
- विद्युत् बलुच्चैरबाट एक एकभेटा अनुकूलनात्मक शब्द, उद्धान र टुक्का पहिचान गरी तिनलाई सक्कमा प्रयोग गर्नुहोस् :
- हरे विषयको चिन्तो पात भने भै सवैलाई पछाडि पारेर सबैलेकुटताको ताज पहिरिएको अनुजको सक्कमाबाट उनको बाबुमा आएको सिमाहीन खुसीको वर्णन मायदै कसैले गर्न सक्ता । अकबरी मुसलाई कमी लगाउनु पर्दैन भन्दै मेघावी विद्यापी अनुजले सवैको नाक राखेकोमा आफूलाई धेरै रुपी लागेको कुरा विद्यालयका प्राचार्यले नुरक्क उठेर मुनुमुनु हान्दै बताए ।
७. कुनै दुई प्रश्नको उत्तर दिनुहोस् । (२+२=४)
 क। तलको बलुच्चैरबाट दुई भेटा उत्पत्त्य शब्द र दुई भेटा प्रत्यक्ष व्यापन्न शब्द पहिचान गरी लेख्नुहोस् :
 शारीरिक प्रामाणिकतालाई महत्त्व दिनुहुन् । अनुमानले मात्रै वास्तविकताको संज्ञान नहोला । मानवीय जीवनमा उपलब्ध अनुभवलाई व्यापक बनाउन सक्दा मात्रै दीर्घायु रहनुको औचित्य हुन्छ ।
 - ख। तलको बलुच्चैरबाट दुई समस्त शब्द र दुई दिव्य शब्द थोडी समस्त शब्दलाई विग्रह गर्नुहोस् र दिव्य भएर बनेका शब्दको दोहोरिएको भाषा लेख्नुहोस् :

- मेरो जग्गाको लागतलाई भावाङ्कित छुट्टेको छ । विमल हुनका लागि म कोसिस गरिरहेको छु । मनले बोधे जस्तो सरासर हुँदै गए त मज्जा हुन्छ्यो । म बासले कम्पर मलाई मलाई लाग्यो र जसरीतसरी धीवारको बन्धनको टाँगे ।
- ग। तलको बलुच्चैरबाट चार भेटा सविशुद्ध व्युत्पन्न शब्द पहिचान गरी तिनको लागि विच्छेद गर्नुहोस् :
 नेपाल धुमन आएका पछिउत्तकहरू नेपालीको स्वागत र काँतिथ पाएर अचानक झर्पित हुँदै डोलीले अनुहारमा आ-आफ्ना मुस्क फर्काउँछन् ।
 - घ। तलका मध्ये कुनै एक प्रश्नको उत्तर दिनुहोस् : (४)
 क। तलको बलुच्चैरलाई पुमिद्वय भए स्त्रीविद्वय भए पुमिद्वयमा परिवर्तन गरी बाक्व ङीथा समेत मिलाई पुनर्निर्माण गर्नुहोस् :
 मेरा बाबा आज बजार जानुभएको छ । भाइले आफूलाई खानेकुरा मगाएको छ । काकाले खेत बन्ने कामान लिएर आउन भन्यो । भिनाजु आज नै हाथो घर आउँदै हुनुहुन्छ ।
 - ख। तलको बलुच्चैरका शब्दलाई बचन परिवर्तन गरी आनन्दक बाक्व ङीथा मिलाई लेख्नुहोस् :
 तिमी बजार जाऊ । तिमी किताब किन । बजारमा भेटिएका साथीहरूसँग परिचय गर । उनीहरूको घर देगाना सोध ।
 - ङ। विद्युत् बलुच्चैरलाई परस्परगत्य मिलाई पुनर्निर्माण गर्नुहोस् :
 मैले आमालाई डोर्गिदियो । आमाको मलाई अशीर्वाद दिइन् । तिमीले पनि आमालाई डोर्गिन् । मेरा भाइले पनि डोर्गो ।
 - च. दुई प्रश्नको उत्तर दिनुहोस् : (४+४=८)
 क। विद्युत् बलुच्चैर पढी लगभग सोधिएका प्रश्नहरूको उत्तर दिनुहोस् :
 आधुनिक भौतिक विज्ञानले विषयमा एउटा नयाँ क्रान्तिको जन्म दिएको छ । भौतिक विज्ञानका प्रमेता न्युटनलाई मानिन्छ । तर न्युटन मात्र नभएर ग्यालिलियो र आइन्स्टाइन यी तीनै जना भौतिक विज्ञानका प्रमेता मानिन्छन् । न्युटनले बालको नियम, ग्यालिलियोले खगोल विज्ञान र आइन्स्टाइनले सापेक्षता (आफू स्वतन्त्र नभएर अर्कोसित सम्बद्ध हुने गुण वा अवस्था) को नियम पत्ता लगाएका हुन् । भौतिक विज्ञानको आरम्भ प्राचीन छिक्काबाट भएको मानिन्छ । यस्तो शक्ति र बलसँग सम्बन्धित समय र स्थानबाट घाल्न र पदार्थको अध्ययन गर्दछ । अर्थात् यस्तो बलको घाल्न र त्यसमा बलको अवरका बारेमा अध्ययन गर्दछ । त्यही मात्र नभएर यस्तो विद्युत् तरङ्ग र त्यसको अन्तरको पनि अध्ययन गर्दछ । विद्युत् तरङ्गकै माध्यमबाट सात समुद्रपारिका मानिससँग प्रत्यक्ष कुराकानी गर्न सकिन्छ । यसको अध्ययनको क्षेत्र असीमित छ । भौतिक विज्ञानकै माध्यमबाट प्रकृति र जीवन जगतका विविध पक्षको अध्ययन गरिन्छ । यस विज्ञानले स्वास्थ्य, कला, गीत, विद्युत्, प्रकाश, ऊर्जा, भूगर्भ तथा खगोल आदि अनेक विषयको प्रामाणिक अध्ययन गर्दछ । समग्रमा प्रकृति, जीवनजगत तथा बहुमाण्डलमेतको बृहत् क्षेत्रलाई यस्तो आफ्नो अध्ययनमा समेटेको छ ।
- प्रश्नहरू :
- (क) भौतिक विज्ञानका प्रमेता को को हुन् ?
 - (ख) भौतिक विज्ञानका आधारमा शक्ति र बलमा के भिन्नता छ ?
 - (ग) भौतिक विज्ञानले कसरी प्रकृतिको व्यापक क्षेत्रलाई समेटेको छ ?
 - (घ) 'भूगर्भ' र 'खगोल' शब्दको अर्थ लेख्नुहोस् ।
- ख। विद्युत् बलुच्चैर पढी सोधिएका बस्तुगत प्रश्नहरूको उत्तर लेख्नुहोस् :
 प्राकृतिक प्रकोप भन्नाले बाढी, पहिरो, वर्षा, खडेरी, हिमपात, हुरीबाटल जस्ता कुराहरू पर्दछन् । प्राकृतिक प्रकोप मानिसहरूको बलमा हुँदैन तर मानिसले चाहेमा यस्तो न्यूनीकरण गर्न सक्दछ, धनजनको क्षतिबाट जोगिन सक्दछ । प्राकृतिक प्रकोपको ख्याल नगर्ने हो भने यस्तो धनजनको विनाश गरी मानव जीवनलाई तहतनहल बनाउँदछ । नेपालको सन्दर्भमा प्राकृतिक प्रकोपमा सबैभन्दा बढी जनमानसलाई दुःख दिने बाढीपहिरो एक हो । वर्षा लागेपछि बाढीपहिरोका कारणले धेरै धनजनको क्षति भएका समाचार हामी सुन्दछौ । यस्तो बाढीपहिरोलाई न्यूनीकरण गर्न बाढी आउने ठाउँमा